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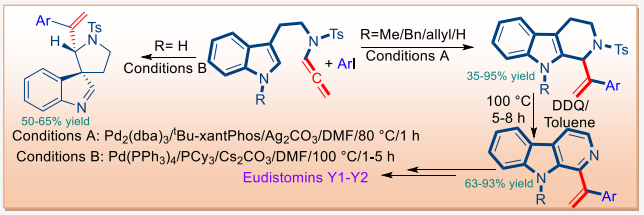
Palladium-Catalyzed Synthesis of 1-Vinyltetrahydro-β-carbolines and Aza-spiroindolenines: Access to the Syntheses of 1-Vinyl-β-carbolines and *Eudistomins Y1* and *Y2*

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ABSTRACT: A general synthesis of 1-vinyltetrahydro-β-carbolines (THBCs) has been achieved via palladium(0)-catalyzed cyclocondensation between allenyltryptamines and aryl iodides. Aza-spiroindolenines could also be accessed from the *N*-unsubstituted indole substrates by simply tweaking the reaction conditions. DDQ-mediated oxidation of THBCs easily afforded β-carbolines, which could be synthetically transformed into 1-aroyl-β-carbolines of pharmacological interest. Formal total syntheses of *eudistomins Y1* and *Y2* have also been achieved.



INTRODUCTION

The tetrahydro-β-carboline (THBC; **1**, Figure 1) and β-carboline (BC; **2**, Figure 1) ring systems, considered as

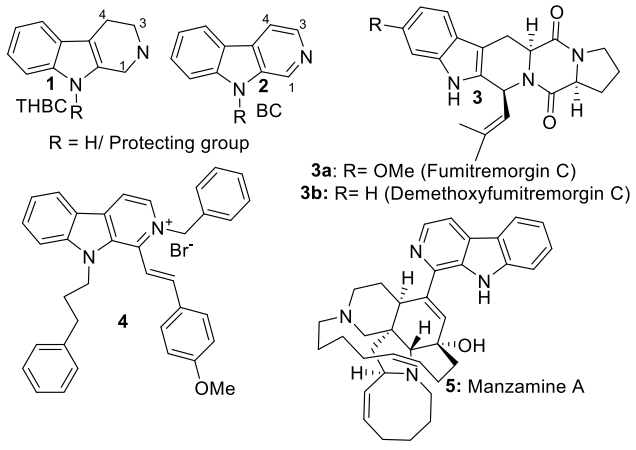


Figure 1. A few biologically active THBC (**3a**, **b**) and β-carboline (**4** and **5**) derivatives.

privileged heterocycles, have attracted substantial attention as synthetic targets because of their presence as core structure in a large number of biologically active and structurally diverse alkaloids and synthetic compounds[1a,f] including pharmaceuticals and their utilization as a popular template in drug discovery programs.[1g] Incorporation of a vinyl moiety at the C1 position of both THBCs and BCs proved to be efficacious in improving the pharmacological effects in a number of compounds. For example, the fungal toxin *fumitremorgin C* (**3a**, Figure 1) proves to be a potent and specific inhibitor of breast cancer resistant protein[2a] (BCRP/ABCG2) multidrug transporter, while the demethoxy analogue **3b** (Figure 1) effectively arrests the cell cycle[2b] (at G2/M phase), indicating its implication in cancer therapy. In the recent past, screening of a number of derivatives of the alkaloid harmine identified the β-carboline derivative **4** as a potent anticancer lead,[2c] while **5** emerged as a highly potent, orally bioavailable antimalarial agent.[2d]

Among the large number of THBCs, C1-substituted derivatives widely exist in nature as a constituent of indole alkaloids; consequently, there have been many reports[3] on their syntheses including those involving modern transition-metal catalysts (Scheme 1). Most of them adopted the well-known Pictet−Spengler (PS) reaction,[4] which however proves less useful when 1-vinyl-substituted THBCs are targeted. Only a few reports[5] exist where *N*-allyl,[5a] allenyl,[5b] or propargyl[5c] derivatives of tryptamine were exposed to rhodium, gold and palladium catalyst respectively (Scheme 1a−c), besides other methods[5d,e] employing oxidative C(sp^3)−H olefination at the C1 position of the preformed THBCs. Consequently, the applications of C1-vinyl THBCs in medicinal chemistry or other fields are limited. Therefore, development of a straightforward and practical method for the general synthesis of C1-vinyl-substituted THBCs remains worthwhile. It is interesting to note that substrates utilized in the formation of THBCs or the compounds resembling the structure of those

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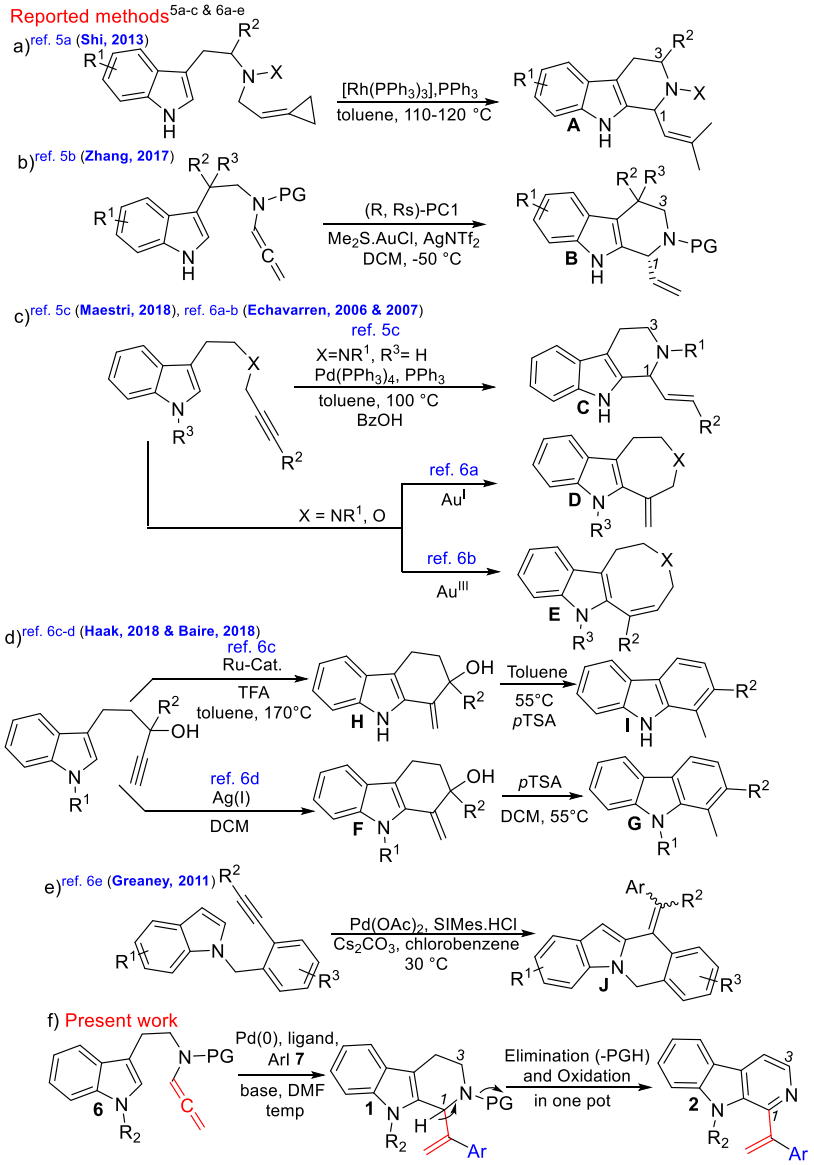


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Scheme 1. Previous Syntheses of THBCs and Related Carbo-/Heterocycles and the Envisaged Synthetic Route for Products 1 and 2



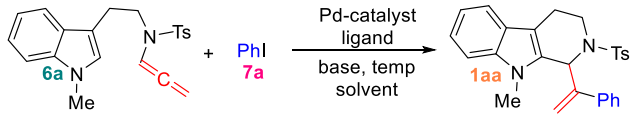
substrates have been the subject of intensified research[6] in this field resulting in the generation of indole fused products **D**–**J** (Scheme 1d,e).

On the other hand, C1-substituted *β*-carbolines have generated considerable synthetic interest[7] in recent times as these are ubiquitous in pharmaceuticals and a number of biologically active compounds. In particular, C-1 vinyl-substituted *β*-carbolines provide easy access to different alkaloids[8] (komaroine, perlolyrine, etc.) and lead compounds[2c] and undergo RCM reactions[9a,b] for the construction of different fused polycyclic scaffolds present in biologically active alkaloids[9a,c,d] (i.e., flavopereirine,[9a] flavocoryline,[9a] cordatanine,[9c] lutonins C and D[9d] etc.). To our surprise, scrutiny of the literature reveals only a few reports[2c,8a,10] on the synthesis of such vinyl-substituted *β*-carbolines though a number of methods are available for other C1-substituted derivatives, achieved primarily via the oxidation of THBCs.[7] More specifically, the installation of the double bond at the C1 position of *β*-carbolines is usually made using palladium-catalyzed coupling[8a] of 1-chloro-*β*-carbolines prepared through multiple steps or a condensation reaction between the derivatives of a harmine alkaloid and an aldehyde.[2c,10] Hence, development of a facile and general method for the construction of 1-vinyl *β*-carbolines from readily available precursors continues to be fascinating.

In continuation of our work[11] on palladium-catalyzed synthesis of heterocycles of biological interest and in view of the immense potential of the *N*-allenamide substrates in recent synthetic transformations,[12] we envisioned that a palladium(0)-catalyzed cyclocondensation between *N*-allenamides **6**, derived readily from *N*-propargylated tryptamine through base (i.e., NaH) treatment, and aryl iodides **7** would afford 1-vinyl THBCs **1** (Scheme 1f). Furthermore, a straightforward base-induced elimination and subsequent in situ oxidation of products **1** could provide a general entry into the class of *β*-carbolines **2** (Scheme 1f). Our hypothesis proved to be viable upon choosing appropriate catalyst and reaction conditions; the results obtained so far are described below.

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Table 1. Optimization of the Reaction Conditions[a]



entry	[Pd]	ligand	base	solvent	temp (°C)	time (h)	yield[b] (%)
1	$Pd(OAc)_2$	PPh_3	K_2CO_3	DMF	80	26	0
2	$Pd(OAc)_2$	PPh_3	Ag_2CO_3	DMF	80	24	15
3	$Pd_2(dba)_3$	PPh_3	Ag_2CO_3	DMF	80	1	26
4	$Pd_2(dba)_3$	*t*-Bu-XPhos	Ag_2CO_3	DMF	80	1	71
5	**$Pd_2(dba)_3$**	***t*-Bu-XantPhos**	**Ag_2CO_3**	**DMF**	**80**	**1**	**75**
6	$Pd_2(dba)_3$	*t*-Bu-XantPhos	Ag_2CO_3	DMF	70	3	70
7	$Pd_2(dba)_3$	Xantphos	Ag_2CO_3	DMF	80	4	30
8	$Pd_2(dba)_3$	*N*-XantPhos	Ag_2CO_3	DMF	80	5	22
9	$Pd(dba)_2$	*t*-Bu-XantPhos	Ag_2CO_3	DMF	80	1	20
10	$Pd(PPh_3)_4$	*t*-Bu-XantPhos	Ag_2CO_3	DMF	80	10	40
11	$Pd_2(dba)_3$	*t*-Bu-XantPhos	Ag_2CO_3	DMSO	80	1	40
12	$Pd_2(dba)_3$	*t*-Bu-XantPhos	Ag_2CO_3	DMA	80	2	29
13	$Pd_2(dba)_3$	*t*-Bu-XantPhos	Ag_2CO_3	MeCN	80	3	38
14	$Pd_2(dba)_3$	*t*-Bu-XantPhos	Ag_2CO_3	toluene	80	1	35

[a]Reaction conditions: **6a** (0.10 mmol), [Pd] (5 mol %), ligand (12 mol %), base (0.2 mmol), DMF (0.03 M). [b]Yield of isolated pure product.

RESULTS AND DISCUSSION

The study was initiated by exploring the model reaction between allenamide **6a** and phenyl iodide **7a** in DMF at 80 °C utilizing 5 mol % of $Pd(OAc)_2$ and 12 mol % of PPh_3 in the presence of 2.0 equiv of potassium carbonate[11d] (Table 1, entry 1); to our disappointment, the desired product **1aa** was not formed even with different organic and inorganic bases (results not shown).

Use of Ag_2CO_3 (2.0 equiv) as base did provide **1aa**, albeit in low yield (Table 1, entry 2); the yield could be improved slightly (Table 1, entry 3) by employing $Pd_2(dba)_3$ as catalyst. To our delight, subsequent reactions (Table 1, entries 4−8) using other ligands showed that the use of the bulky *t*-Bu-XPhos instead of PPh_3 completed the reaction within 1 h, affording the product **1aa** with 71% yield (Table 1, entry 4). Replacement of the phosphine ligand with a bidentate one (i.e., *t*-Bu-XantPhos) improved the yield further to 75% (Table 1, entry 5), but lowering of the reaction temperature to 70 °C entailed longer reaction time (i.e., **3h**) and reduced the yield to some extent (Table 1, entry 6). Surprisingly, use of other bidentate ligands such as Xantphos and *N*-XantPhos had detrimental effects on this reaction, as minor amounts of multiple undesired products were formed, diminishing the yield of **1aa** (Table 1, entries 7 and 8). Use of other palladium(0) catalysts like $Pd(dba)_2$ and $Pd(PPh_3)_4$ separately resulted in moderate yields (Table 1, entries 9 and 10), suggesting $Pd_2(dba)_3$ to be the best. Similarly, employing other solvent systems both polar and nonpolar (Table 1, entries 11−14) furnished the product in moderate yields (29−40%) only. Thus, the reaction conditions of entry 5 of Table 1 were found to be optimal.

We next investigated the substrate scope to examine the generality of this reaction. As can be seen from Table 2, a wide range of allenamides **6** and aryl iodides **7** could easily participate in the reaction, affording THBCs **1** within 1 h in moderate to excellent yields (35−95%). However, use of nosyl (Ns) as a nitrogen protecting group in place of tosyl (Ts) in the allenamide moiety of substrate **6** had a detrimental effect, lowering the yield of products (**1ba**/**1bb**). Gratifyingly, the reactions of **6a** with a bulky 1-iodonaphthalene (**7c**) and 2-iodothiophene (**7d**) were also successful, furnishing the expected products **1ac** and **1ad** though in moderate yields (55−60%). Aryl iodides, **7b** and **7e**, **f** bearing an electron-withdrawing group at the para position performed better than those (**7g**/**7h**) having an electron-donating group (compare the yields of **1ab**/**1ae**/**1af** vs **1ag**/**1ah**). Surprisingly, replacement of the *N*-methyl group (R_2 = Me) in allenamide **6a** by a benzyl or allyl group hindered the reaction to some extent, resulting in lower yields (37−60%) of products (**1ca**/**1ch**/**1db**/**1de**), though *N*-butyl (R_2 = Bu) in allenamide **6i** favored the reaction, furnishing the product **1ia** in very good yield (76%).

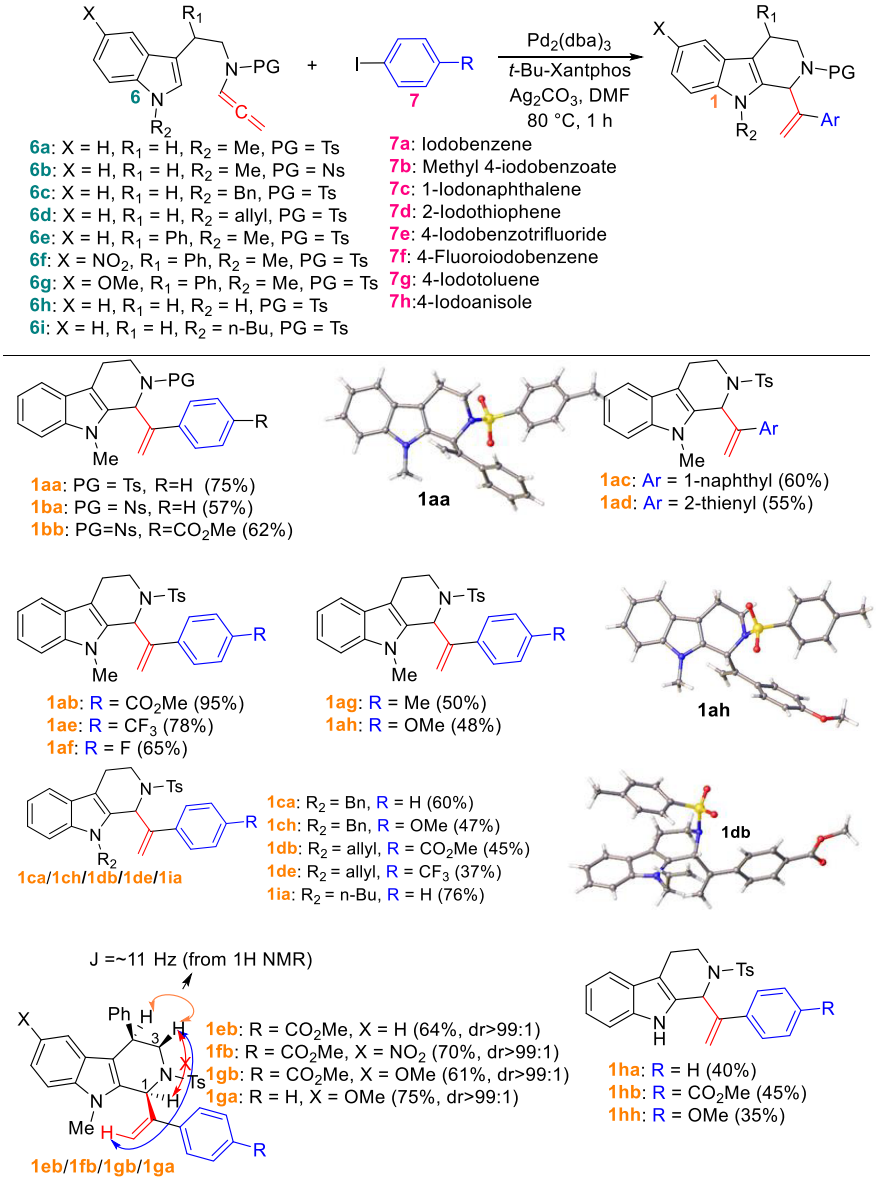
On the other hand, incorporation of a phenyl group in the alkyl chain tethered to the C3 carbon of the indole moiety (R_1 = Ph) as in allenamide **6e** delivered the product **1eb** in good yield (64%) and high diastereoselectivity (dr >99:1); even placement of an additional functional group (X = NO_2/OMe) at C-5 position of the indole worked equally well, affording the desired products (**1fb**/**1gb** and **1ga**, dr >99:1). Substituents at the C1 and C4 positions in THBC ring of products **1eb**, **1fb**, **1gb** and **1ga** (see Table 2) were found to be in a *cis*-relationship, which is in line with a previous report.[4e] This relative stereochemistry was established using NMR data including NOESY experiments (for details, see Figures S6 and S7).

We also explored the prospect of utilizing allenamide **6h** having a free NH group on the indole ring by executing the reactions with iodobenzene (**7a**), methyl 4-iodobenzoate (**7b**), and 4-iodoanisole (**7h**) separately; gratifyingly, the desired products **1ha**, **1hb**, and **1hh** were formed within 1 h albeit in moderate yields (35−45%). Notably, neither the aryl bromide nor the allenamide **6** bearing a sulfonyl group (e.g., R_2 = Ts) at the nitrogen atom of the indole ring was found to be reactive.

In addition, we became interested in checking the prospects of acid-catalyzed cyclization of allenamide substrates **6**.[13] To our pleasure, overnight exposure of allenamide **6a** to 10 mol % *p*-TsOH led to the formation of THBC **1a** successfully albeit in moderate yield (Scheme 2).

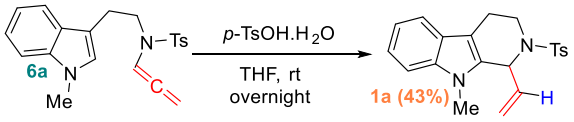
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Table 2. Palladium-Catalyzed Synthesis of THBCs 1[a–c]



[a]Reaction conditions: allenamide **6** (1.0 equiv) was treated with iodide **7** (1.3 equiv) in the presence of $Pd_2(dba)_3$ (0.05 equiv), *t*-Bu-XantPhos (0.12 equiv), and Ag_2CO_3 (2.0 equiv) in DMF under argon. [b]Yield of isolated pure product. [c]H-3β showed NOE with one olefinic proton, but not with H-1α.

Scheme 2. Transformation of 6a into THBC 1a[a,b]

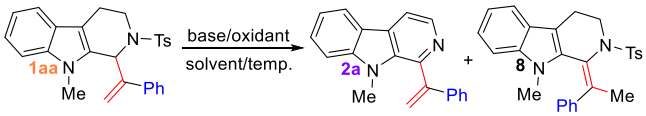


[a]Reaction conditions: allenamide **6a** (1.0 equiv) was treated with *p*-TsOH (10 mol %) in dry THF under argon. [b]Yield of isolated pure product.

Toward the objective of transforming THBCs into 1-vinyl-β-carbolines **2**, we planned to use a suitable base[14] for detosylation with concomitant aromatization (Table 3). Thus, we treated **1aa** with potassium *tert*-butoxide under oxygen atmosphere (balloon pressure); to our disappointment, there was no sign (TLC) of formation any product and the starting compound **1aa** was found to remain unchanged (Table 3, entry 1). We then carried out this reaction in open air; surprisingly, an undesired product **8** was isolated in 40% yield along with a minor amount of desired product **2a** (5%) (Table 3, entry 2). When we carried out this reaction under argon atmosphere, the product **8** was generated predominantly 86% yield (Table 3, entry 3). Even switching to a more polar solvent like DMF or changing the base (viz., KOH, NaH) did not improve the situation much, resulting in the formation of product **8** in varied yields (Table 3, entries 4–6). Pleasingly, employment of DDQ (3.0 equiv) in dry toluene and carrying out the reaction at 100 °C delivered the desired product **2a** in 80% yield (Table 3, entry 7). But decrease of either the temperature (90 °C) or the amount of DDQ (2 equiv) necessitated the use of longer reaction times (Table 3, entries 8 and 9). Furthermore, carrying out this reaction in a more polar

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Table 3. Optimization of the Reaction Conditions to Synthesize β-Carboline 2a[a]



entry	oxidant	base	solvent	temp (°C)	time	product[b] (%) 2a	8
1[c]		KO^{t}Bu	1,4-dioxane	110	24 h	0	0
2[d]		KO^{t}Bu	1,4-dioxane	110	5 h	5	40
3		KO^{t}Bu	1,4-dioxane	110	5 h	0	86
4		KO^{t}Bu	DMF	rt	5 min		10
5		KOH	DMSO	rt	2 h	0	20
6		NaH	DMF	rt	5 h	0	60
7	**DDQ**		**toluene**	**100**	**6 h**	**80**	**0**
8	DDQ		toluene	90	12 h	80	0
9[e]	DDQ		toluene	100	12 h	80	0
10[f]	DDQ		DMF	100	12 h	5	0
11[f]	DDQ		DMSO	100	12 h	5	0
12[f]	DDQ		MeCN	100	12 h	40	0
13[f]	DDQ		1,4-dioxane	100	12 h	35	0

[a]Reaction conditions: THBC **1aa** (1.0 equiv), base (2.0 equiv) or oxidant (3.0 equiv) used under argon atmosphere except entries 1 and 2. [b]Yield of isolated pure product. [c]The reaction was carried out under oxygen atmosphere (balloon pressure). [d]The reaction was carried out under open air. [e]The reaction was executed using DDQ (2.0 equiv). [f]The reaction was incomplete, and substrate **1aa** was recovered (10−34%).

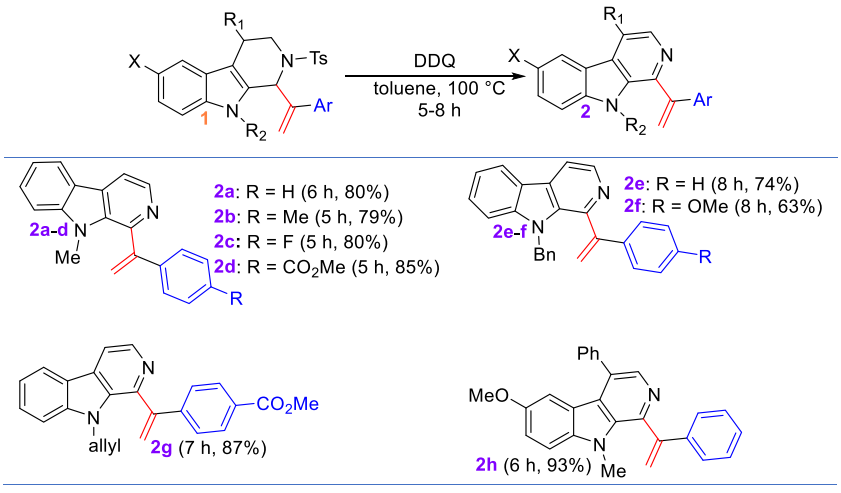
solvent (viz., DMF, DMSO) failed to improve the yield of product **2a**, leaving the starting material mostly unreacted (Table 3, entries 10 and 11), though medium polar solvents, viz. CH_3CN or 1,4-dioxane furnished product **2a** with moderate yields (Table 3, entries 12 and 13). Thus, the reaction conditions of entry 7 of Table 3 proved to be the best for further exploration.

Thereafter, a number of THBCs **1** were exposed to DDQ in order to establish the generality of the protocol (Table 4). The reactions were found to be complete within 5−8 h, and a range of functional groups (e.g., Me, allyl, OMe, CO_2Me, F) were compatible. Nevertheless, an electron-withdrawing group (F/CO_2Me) in the para position of the side chain phenyl ring facilitated this reaction compared to electron-donating (Me/OMe) groups (viz. **2c**/**2d** vs **2b**/**2f**). However, substitution at the indole ring nitrogen had no significant effect (viz. **2a** vs **2e** and **2d** vs **2g**), though an *N*-methyl substituent expedited the reactions compared to an *N*-benzyl or *N*-allyl group. Interestingly, the THBC substrate containing a C5 methoxy group (X = OMe) delivered the β-carboline **2h** with the highest yield (93%).

We then turned our attention to explore the scope of this reaction. To our surprise, tweaking the conditions for the reaction between allenamide **6h** and aryl iodides **7** by replacing $Pd_2(dba)_3$ and Ag_2CO_3 with $Pd(PPh_3)_4$ and Cs_2CO_3, respectively (Table 2), triggered a Friedel−Crafts-type allylic alkylation[15] (at C3 position of indole) resulting in the generation of aza-spiroindolenines **9** (Scheme 3). Initially, we selected four aryl iodides comprising iodobenzene (**7a**), methyl 4-iodobenzoate (**7b**), 4-iodoacetophenone (**7i**), and 4-iodotoluene (**7g**) to obtain the corresponding products **9a** (63%), **9b** (65%), **9i** (61%), and **9g** (56%), respectively, within 1−5 h; the unfavorable role of the electron-donating group in the iodide substrate **7** was noticeable. Furthermore, products **9a** (dr 1.5:1) and **9g** (dr 2.2:1) were obtained as inseparable mixtures of two diastereomers with moderate diastereoselectivity as evident from their ^{1}H NMR spectra (see Figures S8 and S9), while aryl iodides **7b**/**7i** having an EWG group triggered the formation of products **9b**/**9i** with very good diastereoselectivity (dr 9:1). Notably, correlation peaks from NOESY revealed that Ha and He are on the same face of the pyrrolidine ring, as Ha proton signals showed correlation with He signals but not with those of Hf/g (see Figure 2). These NOESY observations (for details, see Figure S10 amd Table S5) clearly suggest that the relative stereochemistries are 3*R* and 11*S*.

The synthesis of more such products **9** is currently underway. Despite the known pharmacological effects of the aza-spiroindolenines,[16] preparations of such compounds have been limited[17] to the $[Ir(cod)Cl]_2$-catalyzed[17a] spirocyclizations of *N*-allyl tryptamine derivatives and [2,4-

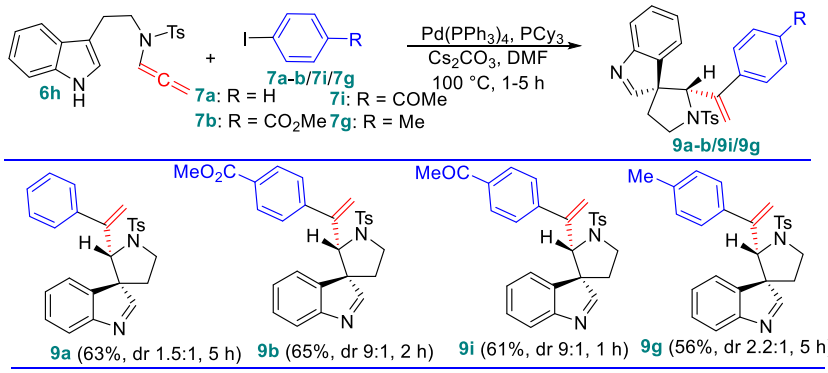
Table 4. Transformations of 1 into β-Carbolines 2[a,b]



[a]Reaction conditions: tetrahydro-β-carboline **1** (1.0 equiv) and DDQ (3.0 equiv) were used. [b]Yield of isolated pure product.

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Scheme 3. Synthesis of Aza-spiroindolenines 9[a,b]



[a]Reaction conditions: allenamide **6h** (1.0 equiv) was treated with iodide **7** (1.1 equiv), $Pd(PPh_3)_4$ (0.05 equiv), PCy_3 (0.12 equiv), and Cs_2CO_3 (4.0 equiv) in dry DMF under argon. [b]The dr was determined by ^1H NMR spectroscopy.

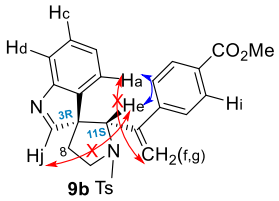
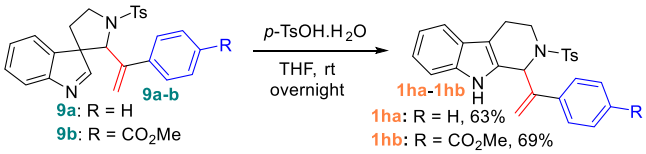


Figure 2. Observed NOESY correlations of compound **9b**.

$(^tBu)_2C_6H_3O]_3PAuCl/AgNTf_2$-catalyzed[17b] intramolecular cyclization of a C2-methyl-substituted derivative of **6h**. Our straightforward and simple reaction protocol could therefore serve as an alternative and useful method for aza-spiroindolenines.

Exposure of a few of our products (**9a**, **9b**) to *p*-TsOH at rt overnight furnished the THBC products (**1ha**, **1hb**) in 63−69% yield (Scheme 4). These results are in line with a previous report[17a] and could be useful in extending the prospect of synthetic transformations of products **9**.

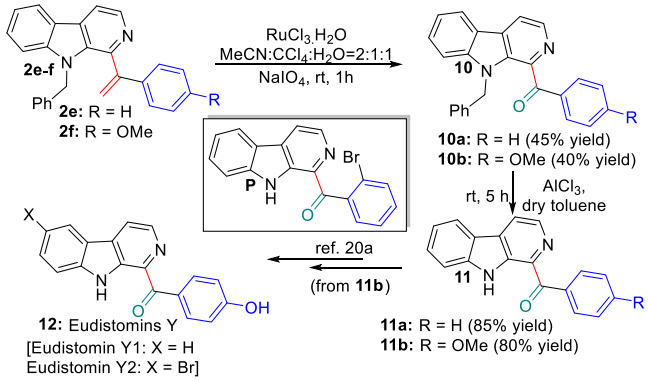
Scheme 4. Transformations of Azaspiroindolenines into THBCs 1[a,b]



[a]Reaction conditions: azaspiroindolenine **9** (1.0 equiv) treated with *p*-TsOH.H_2O (10 mol %) in dry THF under argon. [b]Yield of isolated pure product.

To remove the toxicity of the anticancer agent fascaplycin,[18] a number of *β*-carboline derivatives were designed and synthesized in multiple steps adopting the classical Bischler−Napieralski cyclization and screened against cyclin-dependent kinases[18b] (CDK4 and CDK2). This study resulted in the identification of a compound **P** (Scheme 5) as the most active and selective inhibitor of CDK4. In continuation of our study to identify anticancer lead compound(s),[19] we therefore became interested in attempting the transformations of few of our products **2** into **11** having a structural resemblance with **P**. Gratifyingly, compounds **2e**, **f** could easily be converted into **11a**, **b** by a sequential oxidative cleavage of the exocyclic C=C bond using $RuCl_3$ (cat.)/$NaIO_4$ followed by $AlCl_3$-mediated

Scheme 5. Synthesis of 1-Aroyl-*β*-carbolines 11a,b and Formal Total Synthesis of *Eudistomins Y1* and *Y2*[a,b,c]



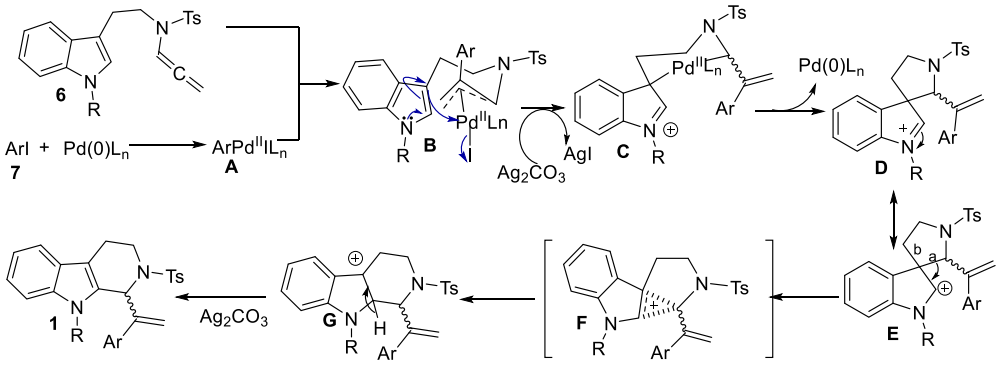
[a]Reaction conditions: A solution containing *β*-carboline **2e-f** (1.0 equiv), $RuCl_3$ (5 mol %), and $NaIO_4$ (6 equiv) in CH_3CN, CCl_4, and H_2O (2:1:1) stirred at rt for 1 h. [b]1-Aroyl-*β*-carboline **10** (1.0 equiv) and anhydrous $AlCl_3$ (3.0 equiv) in dry toluene were stirred at rt for 5 h. [c]Yield of isolated pure product.

debenzylation (Scheme 5). In addition, product **11b** has been reported[20] as an advanced intermediate for the total synthesis of naturally occurring eudistomins[20] Y1 and Y2 (**11**). The anticancer activities (in vitro) of these compounds including **11a**, **b** and the synthesis of more examples of **11** and **12** are currently underway in our laboratories.

On the basis of the experimental results and known palladium chemistry, a plausible reaction mechanism is proposed (Scheme 6). Initially, aryl iodide (ArI) undergoes oxidative addition with Pd(0) to form ArPd(II)I (**A**), which then undergoes addition onto the allenic double bond of **6** resulting in the formation of palladium(II)−*π*-allyl complex **B**.[21] Next, intermediate **B** undergoes intramolecular nucleophilic attack by C3 of the indole moiety with the assistance of Ag_2CO_3[22] triggering the formation of a six-membered palladium(II) species **C**. Reductive elimination of palladium(II) from intermediate **C** would lead to the generation of palladium (0) and a transient dearomatized spiroindolenine intermediate **D** which could undergo allylic group migration[23] from the C3 position of indole to the C2 position possibly via a nonclassical carbonium intermediate **F** (three-centered two-electron bond[24]). The selective formation of intermediate **G** could be attributed to the higher migratory aptitude of allylic

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Scheme 6. Plausible Reaction Mechanism for the Synthesis of Products 1



group than that of methylene group. Finally, deprotonation[25] of **G** would furnish THBC **1**.

Presumably, the silver ion [Ag(I)] might have some role in the allylic group migration (D→G, Scheme 6) which was evident from a control experiment where spiroindolenine **9a** (1 mmol) was heated in DMF (2.5 mL) with Ag_2CO_3 (1.1 mmol); to our pleasure, THBC product **1ha** (for the structure, see Table 2) was isolated in 40% yield though a longer reaction time (24 h) was necessitated.

CONCLUSIONS

In summary, we have developed an expeditious approach for the general synthesis of C1-vinyl-substituted-THBCs **1** via palladium (0)-catalyzed cyclocondensation between *N*-allenyltryptamines **6** and aryl iodides **7**. The reactions are fast, compatible with various functional groups at different sites of the reactants and utilize simple substrates. The THBC products could easily be converted into the corresponding *β*-carbolines **2** using DDQ. Tweaking the reaction conditions for the synthesis of the THBCs **1** could provide a convenient entry to the class of aza-spiroindolenines **9**. Furthermore, synthetic applications of the *β*-carboline products to the 1-aroyl-*β*-carbolines **11a**, **b** of pharmacological interest and the formal total synthesis of eudistomins Y1 and Y2 are also demonstrated. We believe that the method will be useful to practitioners engaged in organic and medicinal chemistry.

EXPERIMENTAL SECTION

General Information. All solvents were distilled using a heating mantle prior to use. Petroleum ether refers to the fraction boiling in the range 60−80 °C. Acetonitrile was dried over phosphorus pentoxide, distilled, and stored over 3 Å molecular sieves in a sealed container. Commercial grade dry DMF (dimethylformamide) and toluene were used as solvents. All reactions were carried out under argon atmosphere and anhydrous conditions unless otherwise noted. Reactions were monitored by thin-layer chromatography (TLC) on silica gel 60 F254 aluminum plates. To visualize the developed chromatogram, we used UV absorbance or iodine exposure. For purification, column chromatography was performed using 100−200 mesh silica gel. ^{1}H and ^{13}C NMR spectra were recorded on a 300, 400 or 600 MHz spectrometer using tetramethylsilane (TMS) as internal standard. Chemical shifts (δ) are given from TMS (δ = 0.00) in parts per million (ppm) with reference to the residual nuclei of the deuterated solvent used [$CDCl_3$: ^{1}H NMR δ = 7.26 ppm (s); ^{13}C NMR δ = 77.0 ppm]. Coupling constants (J) are expressed in Hertz (Hz), and spin multiplicities are given as s (singlet), d (doublet), dd (double doublet), t (triplet), td (triple doublet), m (multiplet), and br (broad). All ^{13}C NMR spectra were obtained with complete proton decoupling. Mass spectra were performed using ESI-TOF and EI ionization mode. Crystallographic data were obtained using Brüker Kuppa Apex 2 instrument.

Procedure for the Synthesis of *N*-Allenamides 6. *N*-Allenamides **6** were synthesized in three steps starting from *N*-protected tryptamine (see Scheme S1). After generation of *N*-propargylated tryptamine (**S3**), allenamides **6** can be derived easily through base (i.e., KO^{t}Bu) treatment from **S3**.

General Procedure for the Synthesis of Acetylenes S2 from S1 (See Scheme S1). The acetylene compound *N*-(2-(1*H*-indol-3-yl)ethyl)-4-methyl-*N*-(prop-2-yn-1-yl)benzenesulfonamide (**S2a**)[5c] was prepared by a known literature procedure. Adopting the same procedure, to a stirred solution of **S1** (10 mmol) in acetone (25 mL) under nitrogen was added K_2CO_3 (2.8 g, 20 mmol), then 3-bromopropyne (1.3 mL, 15 mmol) was added, and the mixture was heated to reflux for 24−48 h. After completion of the reaction (monitored by TLC), water was added to quench the base and then extracted with ethyl acetate. The combined extracts were washed with saturated salt water and dried over anhydrous sodium sulfate. After removal of the solvent under reduced pressure, the crude product was purified by column chromatography over silica gel (100−200 mesh) and eluted with 25−35% ethyl acetate−petroleum ether (v/v) to afford the pure products in 50−98% yield (**S2a-e**).

*N-(2-(1H-Indol-3-yl)ethyl)-4-nitro-N-(prop-2-yn-1-yl)-benzenesulfonamide (**S2b**).*[5b] Yellow solid (3.6 g, 95% yield) isolated using 25% ethyl acetate−petroleum ether (v/v); mp 174−176 °C; ^{1}H NMR ($CDCl_3$, 600 MHz) δ 8.23 (d, J = 9 Hz, 2H), 8.05 (brs, 1H), 7.94 (d, J = 8.4 Hz, 2H), 7.57 (d, J = 8.4 Hz, 1H), 7.35 (d, J = 8.4 Hz, 1H), 7.20 (t, J = 7.5 Hz, 1H), 7.12 (t, J = 7.5 Hz, 1H), 7.08 (d, J = 2.4 Hz, 1H), 4.23 (d, J = 2.4 Hz, 2H), 3.58 (t, J = 7.5 Hz, 2H), 3.10 (t, J = 7.5 Hz, 2H), 2.11 (t, J = 2.4 Hz, 1H); ^{13}C{^{1}H} NMR ($CDCl_3$, 150 MHz) δ 149.9, 144.6, 136.2, 128.7, 127.0, 123.9, 122.3, 122.2, 119.6, 118.4, 111.7, 111.3, 76.2, 74.3, 47.0, 36.6, 24.2 ppm; MS (ESI$^+$) m/z calcd for $C_{19}H_{18}N_3O_4S$ [M + H]$^+$ 384.1018, found 384.1027.

*N-(2-(1H-Indol-3-yl)-2-phenylethyl)-4-methyl-N-(prop-2-yn-1-yl)-benzenesulfonamide (**S2c**).* Brown solid (2.6 g, 60% yield) isolated using 25% ethyl acetate−petroleum ether (v/v); mp 174−176 °C; ^{1}H NMR ($CDCl_3$, 400 MHz) δ 8.19 (brs, 1H), 7.72 (d, J = 8 Hz, 2H), 7.50 (d, J = 7.6 Hz, 1H), 7.37−7.21 (m, 9H), 7.17 (t, J = 7.2 Hz, 1H), 7.05 (t, J = 7.4 Hz, 1H), 4.65 (t, J = 8 Hz, 1H), 4.22−4.15 (m, 1H), 4.08−4.02 (m, 1H), 3.75−3.71 (m, 1H), 3.66−3.61 (m, 1H), 2.40 (s, 3H), 2.06 (d, J = 2.8 Hz, 1H); ^{13}C{^{1}H} NMR ($CDCl_3$, 100 MHz) δ 143.7, 142.0, 136.5, 135.7, 129.6, 128.7, 128.4, 127.9, 127.0, 127.0, 122.7, 122.2, 119.5, 119.2, 115.7, 111.4, 76.6, 74.3, 50.6, 41.5, 36.5, 21.6 ppm; HRMS (ESI$^+$) m/z calcd for $C_{26}H_{25}N_2O_2S$ [M + H]$^+$ 429.1637, found 429.1623.

*4-Methyl-N-(2-(5-nitro-1H-indol-3-yl)-2-phenylethyl)-N-(prop-2-yn-1-yl)benzenesulfonamide (**S2d**).* Yellow gum (3.1 g, 68% yield) isolated using 35% ethyl acetate−petroleum ether (v/v); ^{1}H NMR ($CDCl_3$, 400 MHz) δ 8.58 (brs, 1H), 8.39 (d, J = 2.4 Hz, 1H), 8.05 (dd, J_1 = 9.2 Hz, J_2 = 2 Hz, 1H), 7.68 (d, J = 8.4 Hz, 2H), 7.51 (d, J = 2 Hz, 1H), 7.37 (d, J = 9.2 Hz, 1H), 7.30−7.24 (m, 7H), 4.62 (t, J = 8 Hz, 1H), 4.22 (dd, J_1 = 18.8 Hz, J_2 = 2.4 Hz, 1H), 4.04−3.98 (m, 1H), 3.79 (dd, J_1 = 18.8 Hz, J_2 = 2.4 Hz, 1H), 3.58−3.53 (m, 1H),

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2.39 (s, 3H), 2.12 (t, J = 2.4 Hz, 1H); $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz) δ 143.9, 141.8, 140.8, 139.4, 135.3, 129.7, 129.0, 128.2, 127.8, 127.4, 126.5, 125.9, 118.3, 118.0, 116.5, 111.3, 76.0, 74.8, 50.3, 40.6, 36.3, 21.6 ppm; MS (ESI$^+$) m/z calcd for $C_{26}H_{24}N_3O_4S$ $[M + H]^+$ 474.1488, found 474.1484.

*N-(2-(5-Methoxy-1H-indol-3-yl)-2-phenylethyl)-4-methyl-N-(prop-2-yn-1-yl)benzenesulfonamide (**S2e**).* Brown gum (2.3 g, 50% yield) isolated using 25% ethyl acetate−petroleum ether (v/v); ^1H NMR ($CDCl_3$, 600 MHz) δ 8.01 (brs, 1H), 7.70 (d, J = 8.4 Hz, 2H), 7.36 (d, J = 7.8 Hz, 2H), 7.31 (t, J = 7.5 Hz, 2H), 7.26−7.23 (m, 4H), 7.19 (d, J = 1.8 Hz, 1H), 6.97 (d, J = 2.4 Hz, 1H), 6.84 (dd, J_1 = 9 Hz, J_2 = 2.4 Hz, 1H), 4.58 (t, J = 7.8 Hz, 1H), 4.16 (dd, J_1 = 18.6 Hz, J_2 = 1.8 Hz, 1H), 4.04−4.00 (m, 1H), 3.78 (s, 3H), 3.70 (dd, J_1 = 18.9 Hz, J_2 = 2.1 Hz, 1H), 3.62−3.58 (m, 1H), 2.41 (s, 3H), 2.02 (t, J = 2.1 Hz, 1H); $^{13}C\{^1H\}$ NMR ($CDCl_3$, 150 MHz) δ 153.9, 143.5, 141.7, 135.6, 131.4, 129.4, 128.6, 128.2, 127.8, 127.3, 126.9, 123.2, 115.6, 112.3, 111.8, 101.0, 76.5, 73.9, 55.8, 50.4, 41.7, 36.5, 21.5 ppm; HRMS (ESI$^+$) m/z calcd for $C_{27}H_{26}N_2O_3SNa$ $[M + Na]^+$ 481.1562, found 481.1563.

General Procedure for the Synthesis of Indole-*N*-substituted Acetylenes (S3) (See Scheme S1). The synthesis of 4-methyl-*N*-(2-(1-methyl-1*H*-indol-3-yl)ethyl)-*N*-(prop-2-yn-1-yl)-benzenesulfonamide (**S3a**) has already been reported;[26] we prepared **S3a** following the same reference. Thus, a well-stirred solution of **S2** (1 mmol) in DMF (10.0 mL) was cooled to 0 °C. Then NaH (60% oil suspension; 1.5 mmol) and MeI/benzyl bromide/allyl bromide (1.3 mmol) were added successively to the ice-cold solution of **S2**. Then after 5 min, the reaction mixture was allowed to reach at room temperature gradually and stirred at rt until completion (TLC). After quenching with water (2.0 mL), the mixture was extracted with ethyl acetate (3 × 20 mL). The combined organic extracts were washed consecutively with brine water (10 mL) and dried over anhydrous sodium sulfate. After evaporation of solvent, the residue was purified by silica gel column chromatography with petroleum ether:ethyl acetate = 4:1 (v/v) as eluent to afford the products **S3a−g** in 30−98% yield.

*N-(2-(1-Methyl-1H-indol-3-yl)ethyl)-4-nitro-N-(prop-2-yn-1-yl)-benzenesulfonamide (**S3b**).* Yellow solid (301.7 mg, 76% yield) isolated using 20% ethyl acetate−petroleum ether (v/v); mp 124−125 °C; ^1H NMR ($CDCl_3$, 600 MHz) δ 8.21 (d, J = 9 Hz, 2H), 7.93 (d, J = 9 Hz, 2H), 7.56 (d, J = 7.8 Hz, 1H), 7.28−7.22 (m, 2H), 7.11 (t, J = 7.2 Hz, 1 H), 6.92 (s, 1H), 4.25 (d, J = 2.4 Hz, 2 H), 3.74 (s, 3H), 3.56 (t, J = 7.8 Hz, 2H), 3.08 (t, J = 7.5 Hz, 2H), 2.12 (t, J = 2.4 Hz, 1H); $^{13}C\{^1H\}$ NMR ($CDCl_3$, 150 MHz) δ 149.9, 144.6, 136.9, 128.7, 127.4, 127.0, 123.8, 121.8, 119.1, 118.5, 110.1, 109.4, 76.2, 74.4, 47.2, 36.6, 32.6, 24.1 ppm; HRMS (ESI$^+$) m/z calcd for $C_{20}H_{20}N_3O_4S$ $[M + H]^+$ 398.1175, found 398.1187.

*N-(2-(1-Benzyl-1H-indol-3-yl)ethyl)-4-methyl-N-(prop-2-yn-1-yl)-benzenesulfonamide (**S3c**).* Off-white solid (433.2 mg, 98% yield) isolated using 20% ethyl acetate−petroleum ether (v/v); mp 107−108 °C; ^1H NMR ($CDCl_3$, 400 MHz) δ 7.73 (d, J = 8.4 Hz, 2H), 7.64 (dd, J_1 = 7.6 Hz, J_2 = 0.8 Hz, 1H), 7.37 (d, J = 4.4 Hz, 1H), 7.32−7.24 (m, 5H), 7.19 (td, J_1 = 7.4 Hz, J_2 = 1.2 Hz, 1H), 7.14−7.11 (m, 3H), 7.01 (s, 1H), 5.26 (s, 2H), 4.18 (d, J = 2.4 Hz, 2H), 3.55−3.51 (m, 2H), 3.12−3.08 (m, 2H), 2.40 (s, 3H), 2.08 (t, J = 2.4 Hz, 1H); $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz) δ 143.6, 137.7, 136.7, 136.1, 129.6, 128.9, 128.7, 128.1, 127.8, 127.7, 127.1, 126.9, 126.5, 122.0, 119.3, 119.0, 77.1, 74.0, 50.1, 47.3, 36.9, 24.5, 21.6 ppm; HRMS (ESI$^+$) m/z calcd for $C_{27}H_{27}N_2O_2S$ $[M + H]^+$ 443.1793, found 443.1797.

*N-(2-(1-Allyl-1H-indol-3-yl)ethyl)-4-methyl-N-(prop-2-yn-1-yl)-benzenesulfonamide (**S3d**).* Light yellow solid (345.0 mg, 88% yield) isolated using 20% ethyl acetate−petroleum ether (v/v); mp 64−66 °C; ^1H NMR ($CDCl_3$, 600 MHz) δ 7.73 (d, J = 7.8 Hz, 2H), 7.62 (d, J = 7.8 Hz, 1H), 7.30 (d, J = 8.4 Hz, 1H), 7.27−7.26 (m, 2H), 7.22 (t, J = 7.5 Hz, 1H), 7.12 (t, J = 7.5 Hz, 1H), 6.99 (s, 1H), 6.02−5.95 (m, 1H), 5.21 (d, J = 10.8 Hz, 1H), 5.10 (d, J = 16.8 Hz, 1H), 4.69 (d, J = 5.4 Hz, 2H), 4.20 (d, J = 2.4 Hz, 2H), 3.52 (t, J = 8.1 Hz, 2H), 3.09 (t, J = 7.8 Hz, 2H), 2.41 (s, 3H), 2.09 (t, J = 2.4 Hz, 1H); $^{13}C\{^1H\}$ NMR ($CDCl_3$, 150 MHz) δ 143.4, 136.4, 135.9, 133.5, 129.4, 127.9, 127.7, 125.9, 121.7, 119.1, 118.8, 117.3, 111.1, 109.6, 76.9, 73.7, 48.7, 47.1, 36.7, 24.3, 21.5 ppm; HRMS (ESI$^+$) m/z calcd for $C_{23}H_{25}N_2O_2S$ $[M + H]^+$ 393.1637, found 393.1639.

*4-Methyl-N-(2-(1-methyl-1H-indol-3-yl)-2-phenylethyl)-N-(prop-2-yn-1-yl)benzenesulfonamide (**S3e**).* White crystalline solid (331.5 mg, 75% yield) isolated using 20% ethyl acetate−petroleum ether (v/v); mp 121−123 °C; ^1H NMR ($CDCl_3$, 600 MHz) δ 7.71 (d, J = 7.8 Hz, 2H), 7.49 (d, J = 7.8 Hz, 1H), 7.36 (d, J = 7.2 Hz, 2H), 7.32−7.29 (m, 3H), 7.26−7.20 (m, 4H), 7.09 (s, 1H), 7.04 (t, J = 7.5 Hz, 1H), 4.63 (t, J = 8.1 Hz, 1H), 4.18 (dd, J_1 = 18.9 Hz, J_2 = 2.1 Hz, 1H), 4.01−3.97 (m, 1H), 3.77−3.72 (m, 4H), 3.63−3.60 (m, 1H), 2.41 (s, 3H), 2.05 (t, J = 2.1 Hz, 1H); $^{13}C\{^1H\}$ NMR ($CDCl_3$, 150 MHz) δ 143.4, 141.9, 137.1, 135.6, 129.4, 128.6, 128.2, 127.8, 127.3, 127.1, 126.8, 121.7, 119.2, 118.9, 114.3, 109.2, 76.5, 74.0, 50.4, 41.3, 36.4, 32.8, 21.5 ppm; HRMS (ESI$^+$) m/z calcd for $C_{27}H_{27}N_2O_2S$ $[M + H]^+$ 443.1793, found 443.1792.

*4-Methyl-N-(2-(1-methyl-5-nitro-1H-indol-3-yl)-2-phenylethyl)-N-(prop-2-yn-1-yl)benzenesulfonamide (**S3f**).* Yellow gum (146.1 mg, 30% yield) isolated using 20% ethyl acetate−petroleum ether (v/v); ^1H NMR ($CDCl_3$, 600 MHz) δ 8.44 (d, J = 1.8 Hz, 1H), 8.12 (dd, J_1 = 9 Hz, J_2 = 1.8 Hz, 1H), 7.70 (d, J = 8.4 Hz, 2H), 7.44−7.42 (m, 2H), 7.34−7.33 (m, 5H), 7.28−7.26 (m, 2H), 4.91 (s, 3H), 4.63 (t, J = 8.1 Hz, 1H), 4.22 (dd, J_1 = 18.6 Hz, J_2 = 1.8 Hz, 1H), 4.00−3.96 (m, 1H), 3.81 (dd, J_1 = 18.6, J_2 = 2.4 Hz, 1H), 3.63−3.59 (m, 1H), 2.41 (s, 3H), 2.16 (t, J = 2.1 Hz, 1H); $^{13}C\{^1H\}$ NMR ($CDCl_3$, 150 MHz) δ 143.7, 141.7, 140.5, 138.9, 135.2, 129.6, 128.9, 128.9, 128.1, 127.7, 127.4, 127.3, 118.0, 117.8, 116.7, 109.6, 76.0, 74.7, 50.1, 40.7, 36.5, 36.3, 21.5 ppm; HRMS (ESI$^+$) m/z calcd for $C_{27}H_{25}N_3O_4SNa$ $[M + Na]^+$ 510.1464, found 510.1485.

*N-(2-(5-Methoxy-1-methyl-1H-indol-3-yl)-2-phenylethyl)-4-methyl-N-(prop-2-yn-1-yl)benzenesulfonamide (**S3g**).* White solid (236 mg, 50% yield) isolated using 20% ethyl acetate−petroleum ether (v/v); mp 121−123 °C; ^1H NMR ($CDCl_3$, 600 MHz) δ 7.70 (d, J = 8.4 Hz, 2H), 7.37 (d, J = 7.8 Hz, 2H). 7.31 (t, J = 7.5 Hz, 2H), 7.26−7.23 (m, 3H), 7.17 (d, J = 8.4 Hz, 1H), 7.00 (d, J = 2.4 Hz, 1H), 6.96 (s, 1H), 6.87 (dd, J_1 = 9 Hz, J_2 = 2.4 Hz, 1H), 4.57 (t, J = 8.1 Hz, 1H), 4.14 (dd, J_1 = 18.3 Hz, J_2 = 2.1 Hz, 1H), 4.00−3.95 (m, 1H), 3.79 (s, 3H), 3.72−3.68 (m, 4H), 3.63−3.59 (m, 1H), 2.40 (s, 3H), 2.02 (t, J = 2.4 Hz, 1H); $^{13}C\{^1H\}$ NMR ($CDCl_3$, 150 MHz) δ 153.7, 143.4, 141.8, 135.6, 132.4, 129.4, 128.6, 128.2, 127.8, 127.7, 127.5, 126.8, 113.9, 111.9, 110.0, 101.1, 76.6, 73.9, 55.9, 50.4, 41.7, 36.6, 32.9, 21.5 ppm; HRMS (ESI$^+$) m/z calcd for $C_{28}H_{29}N_2O_3S$ $[M + H]^+$ 473.1899, found 473.1898.

*N-(2-(1-Butyl-1H-indol-3-yl)ethyl)-4-methyl-N-(prop-2-yn-1-yl)-benzenesulfonamide (**S3i**).* Brown gum (367.2 mg, 90% yield) isolated using 20% ethyl acetate−petroleum ether (v/v); ^1H NMR ($CDCl_3$, 400 MHz) δ 7.71 (d, J = 8 Hz, 2H), 7.58 (dd, J_1 = 7.6 Hz, J_2 = 0.8 Hz, 1H), 7.31−7.29 (m, 1H), 7.25−7.17 (m, 3H), 7.08 (td, J_1 = 7.5 Hz, J_2 = 0.9 Hz, 1H), 6.97 (s, 1H), 4.17 (d, J = 2.8 Hz, 2H), 4.05 (t, J = 7 Hz, 2H), 3.51−3.47 (m, 2H), 3.08−3.04 (m, 2H), 2.39 (s, 3H), 2.06 (t, J = 2.4 Hz, 1H), 1.82−1.75 (m, 2H), 1.38−1.28 (m, 2H), 0.93 (t, J = 7.4 Hz, 3H); $^{13}C\{^1H\}$ NMR ($CDCl_3$, 150 MHz) δ 143.5, 136.4, 136.1, 129.5, 127.8, 127.8, 126.0, 121.6, 118.9, 118.9, 110.7, 109.5, 77.0, 73.8, 47.2, 46.0, 36.8, 32.5, 24.4, 21.6, 20.3, 13.8 ppm; HRMS (EI$^+$) m/z calcd for $C_{24}H_{28}N_2O_2S$ $[M]^+$ 408.1872, found 408.1874.

General Procedure for the Synthesis of *N*-Allenamides (6) (See Scheme S1). The *N*-allenamides *N*-(2-(1*H*-indol-3-yl)ethyl)-4-methyl-*N*-(propa-1,2-dien-1-yl)benzenesulfonamide (**6h**)[5c] and 4-methyl-*N*-(2-(1-methyl-1*H*-indol-3-yl)ethyl)-*N*-(propa-1,2-dien-1-yl)-benzene-sulfonamide (**6a**)[26] were prepared by a known literature procedure. To a well-stirred solution of **S3** (0.25 mmol) in dry THF (3 mL) under argon KO^{t}Bu (42 mg, 0.38 mmol) was added an ice-cold condition. The mixture was stirred at room temperature from 5 min to 2 h depending on the starting substrate. After completion of the reaction (checked by TLC), the reaction mixture was quenched with water and then extracted with ethyl acetate. The organic extracts were washed with saturated salt water and dried over anhydrous sodium sulfate. The crude product, which was obtained after removal of solvent under reduced pressure, was purified by column

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chromatography using 15−20% petroleum ether/ethyl acetate (v/v) used as eluting solvent) to obtain the pure products **6** which were characterized by NMR spectroscopy (^{1}H and ^{13}C NMR) and HRMS data. The structure of **6** was further confirmed by the single-crystal X-ray analysis of **6a** (see Figure S4).

Characterization Data for 6. *4-Methyl-N-(2-(1-methyl-1H-indol-3-yl)ethyl)-N-(propa-1,2-dien-1-yl)benzenesulfonamide (**6a**).*[26] Yellow solid (72.3 mg, 79% yield) isolated using 15% ethyl acetate−petroleum ether (v/v); mp 74−75 °C; ^{1}H NMR ($CDCl_3$, 300 MHz) δ 7.68 (d, J = 8.1 Hz, 2H), 7.58 (d, J = 7.8 Hz, 1H), 7.28−7.19 (m, 4H), 7.10 (t, J = 7.4 Hz, 1H), 6.95−6.88 (m, 2H), 5.39 (d, J = 6 Hz, 2H), 3.74 (s, 3H), 3.41−3.36 (m, 2H), 3.05−3.00 (m, 2H), 2.40 (s, 3H); ^{13}C{^{1}H} NMR ($CDCl_3$, 75 MHz) δ 201.4, 143.6, 136.9, 135.5, 129.7, 127.7, 127.1, 127.0, 121.6, 119.0, 118.8, 111.0, 109.3, 100.3, 87.8, 47.5, 32.6, 24.4, 21.6 ppm; HRMS (ESI$^+$) m/z calcd for $C_{21}H_{23}N_2O_2S$ [M + H]$^+$ 367.1480, found 367.1490.

*N-(2-(1-Methyl-1H-indol-3-yl)ethyl)-4-nitro-N-(propa-1,2-dien-1-yl)benzenesulfonamide (**6b**).* Yellow gum (51.4 mg, 52% yield) isolated using 15% ethyl acetate−petroleum ether (v/v); ^{1}H NMR ($CDCl_3$, 600 MHz) δ 8.23 (d, J = 8.4 Hz, 2H), 7.89 (d, J = 8.4 Hz, 2H), 7.52 (d, J = 7.8 Hz, 1H), 7.26−7.24 (m, 1H), 7.21 (t, J = 7.5 Hz, 1H), 7.09 (d, J = 7.5 Hz, 1H), 6.89 (t, J = 6.3 Hz, 1H), 6.86 (s, 1H), 5.44 (d, J = 6 Hz, 2H), 3.73 (s, 3H), 3.48−3.45 (m, 2H), 3.06−3.03 (m, 2H); ^{13}C{^{1}H} NMR ($CDCl_3$, 150 MHz) δ 201.4, 149.9, 144.0, 136.9, 128.1, 127.5, 127.0, 124.1, 121.7, 119.0, 118.7, 110.5, 109.3, 99.4, 88.3, 47.7, 32.6, 24.3 ppm; HRMS (ESI$^+$) m/z calcd for $C_{20}H_{20}N_3O_4S$ [M + H]$^+$ 398.1175, found 398.1184.

*N-(2-(1-Benzyl-1H-indol-3-yl)ethyl)-4-methyl-N-(propa-1,2-dien-1-yl)benzenesulfonamide (**6c**).* Brown gum (88.4 mg, 80% yield) isolated using 20% ethyl acetate−petroleum ether (v/v); ^{1}H NMR ($CDCl_3$, 600 MHz) δ 7.70 (d, J = 8.4 Hz, 2H), 7.61 (d, J = 7.8 Hz, 1H), 7.31−7.24 (m, 6H), 7.17 (t, J = 7.5 Hz, 1H), 7.11−7.09 (m, 3H), 6.96 (s, 1H), 6.92 (t, J = 6 Hz, 1H), 5.35 (d, J = 6 Hz, 2H), 5.27 (s, 2H), 3.43−3.40 (m, 2H), 3.07−3.04 (m, 2H), 2.41 (s, 3H); ^{13}C{^{1}H} NMR ($CDCl_3$, 150 MHz) δ 201.3, 143.6, 137.5, 136.6, 135.6, 129.7, 128.7, 128.7, 128.0, 127.6, 127.1, 126.8, 126.3, 121.8, 119.1, 111.7, 109.7, 100.3, 87.8, 49.9, 47.3, 24.5, 21.5 ppm; HRMS (ESI$^+$) m/z calcd for $C_{27}H_{26}N_2O_2SNa$ [M + Na]$^+$ 465.1613, found 465.1615.

*N-(2-(1-Allyl-1H-indol-3-yl)ethyl)-4-methyl-N-(propa-1,2-dien-1-yl)benzenesulfonamide (**6d**).* Yellow gum (88.2 mg, 90% yield) isolated using 20% ethyl acetate−petroleum ether (v/v); ^{1}H NMR ($CDCl_3$, 600 MHz) δ 7.70 (d, J = 7.8 Hz, 2H), 7.60 (d, J = 7.8 Hz, 1H), 7.29−7.26 (m, 3H), 7.20 (t, J = 7.5 Hz, 1H), 7.11 (t, J = 7.5 Hz, 1H), 6.94−6.92 (m, 2H), 6.01−5.94 (m, 1H), 5.38 (d, J = 6 Hz, 2H), 5.20 (dd, J_1 = 10.5 Hz, J_2 = 1.5 Hz, 1H), 5.10 (dd, J_1 = 17.1 Hz, J_2 = 1.5 Hz, 1H), 4.67 (d, J = 5.4 Hz, 2H), 3.42−3.39 (m, 2H), 3.06−3.03 (m, 2H), 2.41 (s, 3H); ^{13}C{^{1}H} NMR ($CDCl_3$, 150 MHz) δ 201.4, 143.6, 136.3, 135.6, 133.5, 129.7, 127.9, 127.1, 125.9, 121.6, 119.0, 119.0, 117.2, 111.4, 109.6, 100.2, 87.8, 48.6, 47.4, 24.5, 21.5 ppm; HRMS (ESI$^+$) m/z calcd for $C_{23}H_{25}N_2O_2S$ [M + H]$^+$ 393.1637, found 393.1635.

*4-Methyl-N-(2-(1-methyl-1H-indol-3-yl)-2-phenylethyl)-N-(propa-1,2-dien-1-yl)benzene-sulfonamide (**6e**).* Yellow solid (72.9 mg, 66% yield) isolated using 15% ethyl acetate−petroleum ether (v/v); mp 138−140 °C; ^{1}H NMR ($CDCl_3$, 600 MHz) δ 7.64 (d, J = 8.4 Hz, 2H), 7.41 (d, J = 7.8 Hz, 1H), 7.34 (d, J = 7.8 Hz, 2H), 7.30−7.26 (m, 5H), 7.23−7.18 (m, 2H), 7.17 (s, 1H), 7.02 (t, J = 7.5 Hz, 1H), 6.65 (t, J = 6 Hz, 1H), 5.38−5.35 (m, 1H), 5.30−5.28 (m, 1H), 4.75 (t, J = 7.8 Hz, 1H), 3.90−3.87 (m, 1H), 3.78 (s, 3H), 3.56−3.53 (m, 1H), 2.41 (s, 3H); ^{13}C{^{1}H} NMR ($CDCl_3$, 150 MHz) δ 202.5, 143.6, 142.0, 137.1, 134.5, 129.6, 128.4, 128.3, 127.5, 127.4, 127.0, 126.6, 121.6, 119.4, 118.8, 114.6, 109.1, 100.1, 87.6, 51.3, 40.9, 32.8, 21.5 ppm; HRMS (ESI$^+$) m/z calcd for $C_{27}H_{27}N_2O_2S$ [M + H]$^+$ 443.1793, found 443.1806.

*N-(2-(1-Butyl-1H-indol-3-yl)ethyl)-4-methyl-N-(propa-1,2-dien-1-yl)benzenesulfonamide (**6i**).* Brown gum (76.5 mg, 75% yield) isolated using 15% ethyl acetate−petroleum ether (v/v); ^{1}H NMR ($CDCl_3$, 400 MHz) δ 7.67 (d, J = 8.4 Hz, 2H), 7.55 (dd, J_1 = 7.8 Hz, J_2 = 0.6 Hz, 1H), 7.29−7.25 (m, 3H), 7.17 (td, J_1 = 7.6 Hz, J_2 = 0.9 Hz, 1H), 7.06 (td, J_1 = 7.5 Hz, J_2 = 0.9 Hz, 1H), 6.92−6.88 (m, 2H), 5.36 (d, J = 6 Hz, 2H), 4.04 (t, J = 7.2 Hz, 2H), 3.39−3.35 (m, 2H), 3.03−2.99 (m, 2H), 2.39 (s, 3H), 1.79−1.74 (m, 2H), 0.92 (t, J = 7.2 Hz, 3H); ^{13}C{^{1}H} NMR ($CDCl_3$, 100 MHz) δ 201.5, 143.7, 136.3, 135.7, 129.8, 127.9, 127.2, 126.0, 121.5, 119.1, 118.8, 111.0, 109.5, 100.4, 87.8, 47.6, 46.0, 32.4, 24.6, 21.6, 20.3, 13.8 ppm; HRMS (EI$^+$) m/z calcd for $C_{24}H_{28}N_2O_2S$ [M]$^+$ 408.1872, found 408.1869.

General Experimental Procedure for the Synthesis of 1-Vinyltetrahydro-β-carbolines 1 (Table 2). $Pd_2(dba)_3$ (4.6 mg, 5 mol %) and ^{t}Bu-Xantphos (6.0 mg, 12 mol %) were stirred in dry DMF (2 mL) under argon atmosphere at rt for 30 min. Then aryl iodide **7** (0.13 mmol) was added, and stirring was continued for another 30 min at rt. After that, Ag_2CO_3 (55.2 mg, 0.2 mmol) and allenamides **6** (0.1 mmol) were successively added to the reaction mixture under an argon atmosphere. The whole reaction mixture was allowed to stir at 80 °C using an oil bath for 1 h until completion (TLC). The resulting mixture was extracted with DCM (3 × 20 mL) and washed with water (10 mL). The combined organic extracts were dried over Na_2SO_4 and concentrated in vacuo. The crude residue obtained after DMF removal was purified by (silica gel 100−200 mesh) column chromatography using 5−15% ethyl acetate−petroleum ether (v/v) as eluent to afford the desired products **1**. The structures of the productswere fully characterized by NMR spectroscopy (^{1}H and ^{13}C NMR) and HRMS data, confirmed in a few cases by determination of crystal structures.

Procedure for the Gram-Scale Synthesis of 1-Vinyltetrahydro-β-carboline 1aa. A mixture of $Pd_2(dba)_3$ (125.14 mg, 5 mol %) and ^{t}Bu-Xantphos (163.36 mg, 12 mol %) was stirred in dry DMF (55 mL) under argon atmosphere at rt for 30 min. Thereafter, **7a** (0.724 g, 0.40 mL, 3.55 mmol) was added, and stirring was continued for another 30 min at rt. Next, Ag_2CO_3 (1.5 g, 5.46 mmol) and allenamide **6a** (1.0 g, 2.73 mmol) were successively added to the reaction mixture under an argon atmosphere. The whole reaction mixture was allowed to stir at 80 °C using an oil bath for 1 h until completion (TLC). The resulting mixture was extracted with dichloromethane (3 × 30 mL). The combined organic extracts were dried over anhydrous Na_2SO_4, filtered, and concentrated in vacuo. The resulting residue was then purified by column chromatography over silica gel (100−200 mesh) using 5% ethyl acetate−petroleum ether (v/v) as eluent to afford the desired product **1aa** in 65% yield (0.787 g).

Spectral Data for THBCs 1. *9-Methyl-1-(1-phenylvinyl)-2-tosyl-2,3,4,9-tetrahydro-1H-pyrido[3,4-b]indole (**1aa**).* White solid (33.2 mg, 75% yield) isolated using 5% ethyl acetate−petroleum ether (v/v); mp 182−184 °C; ^{1}H NMR ($CDCl_3$, 600 MHz) δ 7.53−7.51 (m, 2H), 7.41−7.38 (m, 4H), 7.32 (d, J = 8.4 Hz, 1H), 7.24−7.22 (m, 3H), 7.10 (t, J = 7.5 Hz, 1H), 6.92 (d, J = 7.8 Hz, 2H), 6.36 (s, 1H), 5.51 (s, 1H), 4.72 (s, 1H), 3.72 (dd, J_1 = 14.4 Hz, J_2 = 6.6 Hz, 1H), 3.68 (s, 3H), 3.37−3.32 (m, 1H), 2.87−2.82 (m, 1H), 2.59 (dd, J_1 = 15.6 Hz, J_2 = 5.4 Hz, 1H), 2.22 (s, 3H); ^{13}C{^{1}H} NMR ($CDCl_3$, 150 MHz) δ 146.7, 143.0, 139.3, 137.0, 131.8, 129.0, 128.5, 128.1, 127.2, 127.1, 126.5, 121.7, 119.7, 119.0, 118.3, 108.8, 108.6, 54.1, 38.5, 29.5, 21.3, 20.1 ppm; HRMS (ESI$^+$) m/z calcd for $C_{27}H_{27}N_2O_2S$ [M + H]$^+$ 443.1793, found 443.1800.

*Methyl 4-(1-(9-Methyl-2-tosyl-2,3,4,9-tetrahydro-1H-pyrido[3,4-b]indol-1-yl)vinyl)benzoate (**1ab**).* Pale yellow solid (47.5 mg, 95% yield) isolated using 10% ethyl acetate−petroleum ether (v/v); mp 150−152 °C; ^{1}H NMR ($CDCl_3$, 600 MHz) δ 8.01 (d, J = 8.4 Hz, 2H), 7.54 (d, J = 8.4 Hz, 2H), 7.38 (d, J = 7.8 Hz, 1H), 7.32 (d, J = 8.4 Hz, 1H), 7.28−7.22 (m, 3H), 7.09 (t, J = 7.5 Hz, 1H), 6.90 (d, J = 8.4 Hz, 2H), 6.29 (s, 1H), 5.57 (s, 1H), 4.81 (s, 1H), 3.96 (s, 3H), 3.81 (dd, J_1 = 14.7 Hz, J_2 = 6.3 Hz, 1H), 3.67 (s, 3H), 3.36−3.31 (m, 1H), 2.80−2.76 (m, 1H), 2.58 (dd, J_1 = 15.9 Hz, J_2 = 5.1 Hz, 1H), 2.21 (s, 3H); ^{13}C{^{1}H} NMR ($CDCl_3$, 150 MHz) δ 166.8, 146.1, 143.8, 143.2, 137.0, 137.0, 131.3, 129.7, 129.6, 129.0, 127.2, 127.0, 126.4, 121.9, 121.1, 119.1, 118.3, 108.8, 108.2, 54.2, 52.2, 38.6, 29.6, 21.2, 20.1 ppm; HRMS (ESI$^+$) m/z calcd for $C_{29}H_{29}N_2O_4S$ [M + H]$^+$ 501.1848, found 501.1852.

*9-Methyl-1-(1-(naphthalen-1-yl)vinyl)-2-tosyl-2,3,4,9-tetrahydro-1H-pyrido[3,4-b]indole (**1ac**).* Pale yellow solid (29.5 mg, 60%

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yield) isolated using 5% ethyl acetate−petroleum ether (v/v); mp 144−146 °C; ^{1}H NMR ($CDCl_3$, 400 MHz) δ 7.93 (d, J = 8 Hz, 1H), 7.82 (d, J = 8 Hz, 1H), 7.77 (d, J = 8 Hz, 1H), 7.48−7.34 (m, 6H), 7.26−7.22 (m, 1H), 7.13−7.09 (m, 1H), 6.91 (d, J = 8.4 Hz, 2H), 6.63 (d, J = 8 Hz, 2H), 6.15 (s, 1H), 5.40 (s, 1H), 5.11 (t, J = 1.2 Hz, 1H), 3.88 (dd, J_1 = 14.4 Hz, J_2 = 6.0 Hz, 1H), 3.81 (s, 3H), 3.55−3.48 (m, 1H), 3.00−2.92 (m, 1H), 2.70 (dd, J_1 = 16 Hz, J_2 = 4.4 Hz, 1H), 2.15 (s, 3H); ^{13}C{^{1}H} NMR ($CDCl_3$, 100 MHz) δ 144.7, 142.5, 137.9, 137.3, 137.0, 134.0, 132.0, 131.7, 128.8, 128.4, 127.9, 126.7, 126.5, 126.2, 125.8, 125.7, 125.2, 125.0, 122.4, 121.9, 119.2, 118.6, 109.3, 109.0, 57.0, 39.2, 30.0, 21.4, 21.1 ppm; HRMS (ESI+) m/z calcd for $C_{31}H_{28}N_2O_2SNa$ [M + Na]$^+$ 515.1769, found 515.1766.

*9-Methyl-1-(1-(thiophene-2-yl)vinyl)-2-tosyl-2,3,4,9-tetrahydro-1H-pyrido[3,4-b]indole (**1ad**).* Pale yellow solid (24.6 mg, 55% yield) isolated using 10% ethyl acetate−petroleum ether (v/v); mp 168−169 °C; ^{1}H NMR ($CDCl_3$, 400 MHz) δ 7.64−7.63 (m, 1H), 7.42−7.39 (m, 2H), 7.30−7.17 (m, 4H), 7.06−7.02 (m, 2H), 6.91 (d, J = 8 Hz, 2H), 6.28 (s, 1H), 5.66 (s, 1H), 4.62 (s, 1H), 3.82 (dd, J_1 = 14.8 Hz, J_2 = 6.4 Hz, 1H), 3.58 (s, 3H), 3.47−3.39 (m, 1H), 2.70−2.61 (m, 1H), 2.53−2.47 (m, 1H), 2.16 (s, 3H); ^{13}C{^{1}H} NMR ($CDCl_3$, 100 MHz) δ 143.4, 142.1, 139.4, 137.1, 137.0, 131.3, 129.0, 128.0, 127.2, 126.4, 126.0, 125.1, 121.9, 119.1, 118.4, 118.3, 108.9, 108.7, 54.3, 38.9, 29.5, 21.3, 19.7 ppm; HRMS (ESI$^+$) m/z calcd for $C_{25}H_{24}N_2O_2S_2Na$ [M + Na]$^+$ 471.1177, found 471.1178.

*9-Methyl-2-tosyl-1-(1-(4-(trifluoromethyl)phenyl)vinyl)-2,3,4,9-tetrahydro-1H-pyrido[3,4-b]indole (**1ae**).* White solid (39.8 mg, 78% yield) isolated using 5% ethyl acetate−petroleum ether (v/v); mp 166−167 °C; ^{1}H NMR ($CDCl_3$, 400 MHz) δ 7.61−7.57 (m, 4H), 7.38 (d, J = 8 Hz, 1H), 7.31 (d, J = 8 Hz, 1H), 7.25−7.21 (m, 3H), 7.11−7.07 (m, 1H), 6.90 (d, J = 8 Hz, 2H), 6.28 (s, 1H), 5.55 (s, 1H), 4.80 (s, 1H), 3.81 (dd, J_1 = 14.8 Hz, J_2 = 6.4 Hz, 1H), 3.66 (s, 3H), 3.37−3.29 (m, 1H), 2.85−2.76 (m, 1H), 2.58 (dd, J_1 = 15.8 Hz, J_2 = 5 Hz, 1H), 2.20 (s, 3H); ^{13}C{^{1}H} NMR ($CDCl_3$, 150 MHz) δ 145.8, 143.3, 142.9, 137.0, 136.9, 131.2, 130.1 (q, J = 32.3 Hz), 129.0, 127.6, 126.9, 126.4, 125.4 (q, J = 3.7 Hz), 124.1 (q, J = 270.3 Hz), 121.9, 121.3, 119.2, 118.4, 108.9, 108.8, 54.2, 38.7, 29.5, 21.2, 20.1 ppm; HRMS (ESI$^+$) m/z calcd for $C_{28}H_{25}N_2O_2F_3SNa$ [M + Na]$^+$ 533.1487, found 533.1479.

*1-(1-(4-Fluorophenyl)vinyl)-9-methyl-2-tosyl-2,3,4,9-tetrahydro-1H-pyrido[3,4-b]indole (**1af**).* Off white solid (29.9 mg, 65% yield) isolated using 10% ethyl acetate−petroleum ether (v/v); mp 141−143 °C; ^{1}H NMR ($CDCl_3$, 400 MHz) δ 7.47−7.43 (m, 2H), 7.36−7.34 (m, 1H), 7.30−7.19 (m, 4H), 7.08−7.01 (m, 3H), 6.91 (d, J = 8.4 Hz, 2H), 6.23 (s, 1H), 5.45 (s, 1H), 4.69 (s, 1H), 3.76 (dd, J_1 = 14.8 Hz, J_2 = 6 Hz, 1H), 3.64 (s, 3H), 3.35−3.27 (m, 1H), 2.79−2.70 (m, 1H), 2.54 (dd, J_1 = 15.6 Hz, J_2 = 5.2 Hz, 1H), 2.20 (s, 3H); ^{13}C{^{1}H} NMR ($CDCl_3$, 100 MHz) δ 162.8 (d, J = 245.9 Hz), 145.7, 143.4, 137.1, 137.1, 135.3 (d, J = 3.4 Hz), 131.6, 129.1, 129.0 (d, J = 8.1 Hz), 127.1, 126.5, 121.9, 120.0, 119.2, 118.4, 115.4 (d, J = 21.2 Hz), 108.9, 108.7, 54.4, 38.7, 29.7, 21.4, 20.1 ppm; HRMS (ESI$^+$) m/z calcd for $C_{27}H_{26}N_2O_2FS$ [M + H]$^+$ 461.1699, found 461.1700.

*9-Methyl-1-(1-(p-tolyl)vinyl)-2-tosyl-2,3,4,9-tetrahydro-1H-pyrido[3,4-b]indole (**1ag**).* White solid (22.8 mg, 50%) isolated using 5% ethyl acetate−petroleum ether (v/v); mp 152−154 °C; ^{1}H NMR ($CDCl_3$, 600 MHz) δ 7.41−7.37 (m, 3H), 7.31 (d, J = 7.8 Hz, 1H), 7.26−7.21 (m, 3H), 7.18 (d, J = 8.4 Hz, 2H), 7.10−7.07 (m, 1H), 6.91 (d, J = 7.8 Hz, 2H), 6.31 (s, 1H), 5.48 (s, 1H), 4.67 (s, 1H), 3.73 (dd, J_1 = 14.4 Hz, J_2 = 6.6 Hz, 1H), 3.66 (s, 3H), 3.37−3.32 (m, 1H), 2.83−2.78 (m, 1H), 2.57 (dd, J_1 = 15.6 Hz, J_2 = 4.8 Hz, 1H), 2.40 (s, 3H), 2.22 (s, 3H); ^{13}C{^{1}H} NMR ($CDCl_3$, 150 MHz) δ 146.5, 142.9, 137.9, 137.1, 137.0, 136.4, 132.0, 129.1, 128.9, 127.1, 127.1, 126.5, 121.6, 119.1, 119.0, 118.2, 108.8, 108.5, 54.2, 38.5, 29.5, 21.3, 21.2, 20.1 ppm; HRMS (ESI$^+$) m/z calcd for $C_{28}H_{29}N_2O_2S$ [M + H]$^+$ 457.1950, found 457.1948.

*1-(1-(4-Methoxyphenyl)vinyl)-9-methyl-2-tosyl-2,3,4,9-tetrahydro-1H-pyrido[3,4-b]indole (**1ah**).* Colorless solid (22.7 mg, 48%) isolated using 10% ethyl acetate−petroleum ether (v/v); mp 120−122 °C; ^{1}H NMR ($CDCl_3$, 400 MHz) δ 7.45−7.42 (m, 2H), 7.35−7.33 (m, 1H), 7.29−7.17 (m, 4H), 7.07−7.03 (m, 1H), 6.90−6.86 (m, 4H), 6.27 (s, 1H), 5.43 (s, 1H), 4.63 (s, 1H), 3.83 (s, 3H), 3.72 (dd, J_1 = 15 Hz, J_2 = 6.6 Hz, 1H), 3.63 (s, 3H), 3.37−3.29 (m, 1H), 2.80−2.72 (m, 1H), 2.54 (dd, J_1 = 15.8 Hz, J_2 = 5 Hz, 1H), 2.19 (s, 3H); ^{13}C{^{1}H} NMR ($CDCl_3$, 100 MHz) δ 159.7, 146.0, 143.1, 137.3, 137.1, 132.1, 131.7, 129.0, 128.4, 127.2, 126.6, 121.7, 119.1, 118.6, 118.3, 113.9, 108.9, 108.6, 55.4, 54.3, 38.6, 29.6, 21.4, 20.2 ppm; HRMS (EI$^+$) m/z calcd for $C_{28}H_{29}N_2O_3S$ [M + H]$^+$ 473.1899, found 473.1896.

*9-Methyl-2-((4-nitrophenyl)sulfonyl)-1-(1-phenylvinyl)-2,3,4,9-tetrahydro-1H-pyrido[3,4-b]indole (**1ba**).* Orange solid (27.0 mg, 57% yield) isolated using 10% ethyl acetate−petroleum ether (v/v); mp 68−70 °C; ^{1}H NMR ($CDCl_3$, 600 MHz) δ 7.93 (d, J = 8.4 Hz, 2H), 7.45−7.42 (m, 5H), 7.39−7.33 (m, 4H), 7.26−7.24 (m, 1H), 7.11 (t. J = 7.5 Hz, 1H), 6.36 (s, 1H), 5.51 (s, 1H), 4.75 (s, 1H), 3.79 (dd, J_1 = 14.4 Hz, J_2 = 6 Hz, 1H), 3.69 (s, 3H), 3.45−3.39 (m, 1H), 3.00−2.94 (m, 1H), 2.72 (dd, J_1 = 15.6 Hz, J_2 = 4.8 Hz, 1H); ^{13}C{^{1}H} NMR ($CDCl_3$, 150 MHz) δ 149.4, 146.2, 146.0, 139.3, 137.0, 131.3, 128.7, 128.4, 128.0, 127.1, 126.2, 123.5, 122.2, 120.4, 119.4, 118.3, 109.0, 108.6, 54.6, 39.0, 29.7, 20.8 ppm; HRMS (ESI$^+$) m/z calcd for $C_{26}H_{24}N_3O_4S$ [M + H]$^+$ 474.1488, found 474.1498.

*Methyl 4-(1-(9-methyl-2-((4-nitrophenyl)sulfonyl)-2,3,4,9-tetrahydro-1H-pyrido[3,4-b]indol-1-yl)vinyl)benzoate (**1bb**).* Yellow solid (32.9 mg, 62% yield) isolated using 10% ethyl acetate−petroleum ether (v/v); mp 204−206 °C; ^{1}H NMR ($CDCl_3$, 600 MHz) δ 7.96−7.91 (m, 4H), 7.52 (d, J = 9 Hz, 2H), 7.47 (d, J = 7.8 Hz, 2H), 7.40 (d, J = 7.8 Hz, 1H), 7.34 (d, J = 7.8 Hz, 1H), 7.27−7.25 (m, 1H), 7.11 (t, J = 7.5 Hz, 1H), 6.27 (s, 1H), 5.59 (s, 1H), 4.84 (s, 1H), 3.96−3.92 (m, 4H), 3.69 (s, 3H), 3.48−3.42 (m, 1H), 2.91−2.85 (m, 1H), 2.71 (dd, J_1 = 15.6 Hz, J_2 = 4.8 Hz, 1H); ^{13}C{^{1}H} NMR ($CDCl_3$, 150 MHz) δ 166.4, 149.4, 146.1, 145.6, 143.5, 137.0, 130.8, 130.0, 129.8, 127.8, 127.1, 126.1, 123.5, 122.4, 121.7, 119.5, 118.3, 109.0, 108.7, 55.0, 52.3, 39.3, 29.6, 20.7 ppm; HRMS (ESI$^+$) m/z calcd for $C_{28}H_{26}N_3O_6S$ [M + H]$^+$ 532.1542, found 532.1545.

*9-Benzyl-1-(1-phenylvinyl)-2-tosyl-2,3,4,9-tetrahydro-1H-pyrido-[3,4-b]indole (**1ca**).* Pale yellow solid (31.1 mg, 60% yield) isolated using 10% ethyl acetate−petroleum ether (v/v); mp 139−141 °C; ^{1}H NMR ($CDCl_3$, 400 MHz) δ 7.42 (d, J = 8 Hz, 1H), 7.38−7.25 (m, 9H), 7.20−7.16 (m, 1H), 7.12−7.07 (m, 5H), 6.86 (d, J = 8.4 Hz, 2H), 6.06 (s, 1H), 5.46 (s, 1H), 5.38 (d, J = 17.2 Hz, 1H), 5.11 (d, J = 16.8 Hz, 1H), 4.75 (s, 1H), 3.83 (dd, J_1 = 14.6 Hz, J_2 = 6.2 Hz, 1H), 3.43−3.35 (m, 1H), 2.82−2.73 (m, 1H), 2.60 (dd, J_1 = 15.8 Hz, J_2 = 5 Hz, 1H), 2.23 (s, 3H); ^{13}C{^{1}H} NMR ($CDCl_3$, 100 MHz) δ 146.8, 143.0, 139.3, 137.5, 137.2, 137.1, 131.8, 129.1, 129.1, 128.5, 128.0, 127.9, 127.3, 127.3, 126.7, 126.6, 122.2, 119.9, 119.4, 118.6, 109.8, 109.5, 54.5, 47.0, 38.4, 21.5, 20.2 ppm; HRMS (ESI$^+$) m/z calcd for $C_{33}H_{31}N_2O_2S$ [M + H]$^+$ 519.2106, found 519.2109.

*9-Benzyl-1-(1-(4-methoxyphenyl)vinyl)-2-tosyl-2,3,4,9-tetrahydro-1H-pyrido[3,4-b]indole (**1ch**).* Pale yellow solid (25.8 mg, 47% yield) isolated using 10% ethyl acetate−petroleum ether (v/v); mp 62−64 °C; ^{1}H NMR ($CDCl_3$, 400 MHz) δ 7.41 (d, J = 7.6 Hz, 1H), 7.38−7.25 (m, 5H), 7.20−7.16 (m, 1H), 7.13−7.07 (m, 6H), 6.87 (d, J = 8.4 Hz, 2H), 6.82 (d, J = 8.8 Hz, 2H), 6.04 (s, 1H), 5.42 (s, 1H), 5.37 (d, J = 16.8 Hz, 1H), 5.09 (d, J = 16.8 Hz, 1H), 4.68 (s, 1H), 3.88−3.82 (m, 4H), 3.44−3.36 (m, 1H), 2.75−2.69 (m, 1H), 2.60−2.55 (m, 1H), 2.23 (s, 3H); ^{13}C{^{1}H} NMR ($CDCl_3$, 100 MHz) δ 159.6, 146.0, 143.0, 137.6, 137.4, 137.1, 132.0, 131.7, 129.1, 129.0, 128.4, 127.3, 126.7, 126.6, 122.1, 119.3, 118.7, 118.5, 113.8, 109.7, 109.5, 55.4, 54.6, 47.0, 38.5, 21.4, 20.2 ppm; HRMS (ESI$^+$) m/z calcd for $C_{34}H_{33}N_2O_3S$ [M + H]$^+$ 549.2212, found 549.2231.

*Methyl 4-(1-(9-Allyl-2-tosyl-2,3,4,9-tetrahydro-1H-pyrido[3,4-b]-indol-1-yl)vinyl)benzoate (**1db**).* Off-white solid (23.7 mg, 45% yield) isolated using 15% ethyl acetate−petroleum ether (v/v); mp 152−154 °C; ^{1}H NMR ($CDCl_3$, 600 MHz) δ 8.00 (d, J = 7.8 Hz, 2H), 7.49 (d, J = 8.4 Hz, 2H), 7.40 (d, J = 7.8 Hz, 1H), 7.32−7.30 (m, 3H), 7.22 (t, J = 7.5 Hz, 1H), 7.10 (t, J = 7.8 Hz, 1H), 6.92 (d, J = 8.4 Hz, 2H), 6.23 (s, 1H), 6.04−5.98 (m, 1H), 5.55 (s, 1H), 5.28 (d, J = 10.2 Hz, 1H), 5.06 (d, J = 16.2 Hz, 1H), 4.85 (s, 1H), 4.76 (dd, J_1 = 16.2 Hz, J_2 = 3 Hz, 1H), 4.56 (dd, J_1 = 17.4 Hz, J_2 = 5.4 Hz, 1H), 3.96 (s, 3H), 3.88 (dd, J_1 = 14.4 Hz, J_2 = 6 Hz, 1H), 3.41−3.36 (m, 1H), 2.80−2.74 (m, 1H), 2.60 (dd, J_1 = 15.6 Hz, J_2 = 4.8 Hz, 1H), 2.24 (s, 3H); ^{13}C{^{1}H} NMR ($CDCl_3$, 150 MHz) δ 166.8, 146.2,

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143.9, 143.2, 137.1, 136.6, 133.2, 131.0, 129.7, 129.6, 129.0, 127.1, 127.1, 126.5, 122.0, 121.1, 119.3, 118.4, 117.2, 109.4, 109.4, 54.2, 52.1, 45.7, 38.5, 21.3, 20.1 ppm; HRMS (ESI^+) m/z calcd for $C_{31}H_{31}N_2O_4S$ $[M + H]^+$ 527.2005, found 527.2007.

*9-Allyl-2-tosyl-1-(1-(4-(trifluoromethyl)phenyl)vinyl)-2,3,4,9-tetrahydro-1H-pyrido[3,4-b]indole (**1de**).* Yellow gum (19.8 mg, 37% yield) isolated using 10% ethyl acetate−petroleum ether (v/v);^{1}H NMR ($CDCl_3$, 400 MHz) δ 7.56 (d, J = 8.4 Hz, 2H), 7.51 (d, J = 8 Hz, 2H), 7.38 (d, J = 8.4 Hz, 1H), 7.29−7.18 (m, 4H), 7.10−7.06 (m, 1H), 6.90 (d, J = 8 Hz, 2H), 6.21 (s, 1H), 6.02−5.95 (m, 1H), 5.52 (s, 1H), 5.26 (dd, J_1 = 10.4 Hz, J_2 = 0.8 Hz, 1H), 5.04 (dd, J_1 = 17.2 Hz, J_2 = 1.2 Hz, 1H), 4.83 (s, 1H), 4.76−4.71 (m, 1H), 4.56−4.50 (m, 1H), 3.89−3.83 (m, 1H), 3.39−3.31 (m, 1H), 2.81−2.73 (m, 1H), 2.58 (dd, J_1 = 15.8 Hz, J_2 = 5 Hz, 1H), 2.22 (s, 3H); $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz) δ 146.0, 143.4, 137.2, 136.7, 133.3, 131.0, 130.1 (q, J = 32.3 Hz), 129.1, 128.5, 127.6, 127.1, 125.4 (q, J = 3.8 Hz), 124.2 (q, J = 270.8 Hz), 122.2, 121.4, 120.1, 119.4, 118.6, 117.3, 109.6, 109.5, 54.3, 45.9, 38.6, 21.4, 20.3 ppm; HRMS (ESI^+) m/z calcd for $C_{30}H_{27}N_2O_2F_3SNa$ $[M + Na]^+$ 559.1643, found 559.1647.

*Methyl 4-(1-(9-Methyl-4-phenyl-2-tosyl-2,3,4,9-tetrahydro-1H-pyrido[3,4-b]indol-1-yl)vinyl)benzoate (**1eb**).* Off-white solid (36.9 mg, 64% yield) isolated using 10% ethyl acetate−petroleum ether (v/v); mp 180−182 °C; ^{1}H NMR ($CDCl_3$, 600 MHz) δ 7.99 (d, J = 8.4 Hz, 2H), 7.51 (d, J = 8.4 Hz, 2H), 7.34 (d, J = 7.8 Hz, 1H), 7.30−7.25 (m, 5H), 7.19−7.17 (m, 3H), 6.92 (d, J = 7.8 Hz, 2H), 6.87 (t, J = 7.5 Hz, 1H), 6.71 (d, J = 7.8 Hz, 1H), 6.39 (s, 1H), 5.65 (s, 1H), 5.08 (s, 1H), 4.34−4.31 (m, 1H), 3.96 (s, 3H), 3.82−3.78 (m, 1H), 3.76 (s, 3H), 3.14−3.09 (m, 1H), 2.22 (s, 3H); $^{13}C\{^1H\}$ NMR ($CDCl_3$, 150 MHz) δ 166.7, 146.6, 143.8, 143.4, 141.6, 137.2, 136.7, 132.6, 129.7, 129.7, 129.1, 128.6, 128.3, 127.2, 127.0, 127.0, 125.8, 121.8, 121.2, 120.1, 119.2, 111.7, 108.8, 54.1, 52.2, 47.2, 39.5, 29.6, 21.2 ppm; HRMS (ESI^+) m/z calcd for $C_{35}H_{32}N_2O_4SNa$ $[M + Na]^+$ 599.1980, found 599.1982.

*Methyl 4-(1-(9-Methyl-6-nitro-4-phenyl-2-tosyl-2,3,4,9-tetrahydro-1H-pyrido[3,4-b]indol-1-yl)vinyl)benzoate (**1fb**).* Pale yellow gum (43.5 mg, 70% yield) isolated using 15% ethyl acetate−petroleum ether (v/v); ^{1}H NMR ($CDCl_3$, 600 MHz) δ 8.10 (dd, J_1 = 9.0 Hz, J_2 = 2.4 Hz, 1H), 8.00 (d, J = 8.4 Hz, 2H), 7.65−7.65 (m, 1H), 7.50 (d, J = 8.4 Hz, 2H), 7.37−7.16 (m, 8H), 6.94 (d, J = 7.8 Hz, 2H), 6.39 (s, 1H), 5.69 (s, 1H), 5.03 (s, 1H), 4.41−4.38 (m, 1H), 3.97 (s, 3H), 3.82−3.78 (m, 4H), 3.16−3.12 (m, 1H), 2.24 (s, 3H); $^{13}C\{^1H\}$ NMR ($CDCl_3$, 150 MHz) δ 166.6, 146.2, 143.7, 143.3, 141.4, 140.4, 140.0, 136.3, 136.2, 130.0, 129.8, 129.2, 129.0, 128.0, 127.7, 127.2, 127.1, 125.2, 121.1, 117.7, 117.0, 114.8, 108.8, 53.8, 52.2, 46.8, 39.3, 29.7, 21.3 ppm; HRMS (ESI^+) m/z calcd for $C_{35}H_{32}N_3O_6S$ $[M + H]^+$ 622.2012, found 622.2012.

*6-Methoxy-9-methyl-4-phenyl-1-(1-phenylvinyl)-2-tosyl-2,3,4,9-tetrahydro-1H-pyrido[3,4-b]indole (**1ga**).* Pale yellow solid (41.1 mg, 75% yield) isolated using 10% ethyl acetate−petroleum ether (v/v); mp 144−146 °C;^{1}H NMR ($CDCl_3$, 400 MHz) δ 7.44−7.42 (m, 2H), 7.35−7.33 (m, 3H), 7.28−7.16 (m, 8H), 6.92 (d, J = 8.4 Hz, 2H), 6.80 (dd, J_1 = 8.8 Hz, J_2 = 2.4 Hz, 1H), 6.40 (s, 1H), 6.11 (d, J = 2.4 Hz, 1H), 5.55 (s, 1H), 4.97 (s, 1H), 4.34−4.30 (m, 1H), 3.70−3.63 (m, 4H), 3.52 (s, 3H), 3.11−3.04 (m, 1H), 2.23 (s, 3H); $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz) δ 153.6, 147.3, 143.2, 141.7, 139.5, 136.9, 133.7, 132.7, 129.1, 128.7, 128.6, 128.5, 128.2, 127.4, 127.3, 127.0, 126.4, 120.0, 111.5, 111.3, 109.5, 102.5, 55.8, 54.2, 47.1, 39.7, 29.9, 21.4 ppm; HRMS (ESI^+) m/z calcd for $C_{34}H_{33}N_2O_3S$ $[M + H]^+$ 549.2212, found 549.2196.

*Methyl 4-(1-(6-Methoxy-9-methyl-4-phenyl-2-tosyl-2,3,4,9-tetrahydro-1H-pyrido[3,4-b]indol-1-yl)vinyl)benzoate (**1gb**).* Brown solid (37.0 mg, 61% yield) isolated using 10% ethyl acetate−petroleum ether (v/v); mp 174−176 °C; ^{1}H NMR ($CDCl_3$, 400 MHz) δ 7.96 (td, J_1 = 8.4 Hz, J_2 = 1.9 Hz, 2H), 7.45 (td, J_1 = 8.4 Hz, J_2 = 1.8 Hz, 2H), 7.28−7.15 (m, 8H), 6.91 (d, J = 8.0 Hz, 2H), 6.81 (dd, J_1 = 8.8 Hz, J_2 = 2.4 Hz, 1H), 6.33 (s, 1H), 6.10 (d, J = 2.4 Hz, 1H), 5.61 (s, 1H), 5.04 (s, 1H), 4.29−4.25 (m, 1H), 3.94 (s, 3H), 3.79−3.74 (m, 1H), 3.69 (s, 3H), 3.52 (s, 3H), 3.11−3.05 (m, 1H), 2.22 (s, 3H); $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz) δ 166.8, 153.7, 146.7, 144.0, 143.4, 141.5, 136.9, 133.1, 132.7, 129.8, 129.8, 129.2, 128.7, 128.4, 127.3, 127.1, 127.1, 126.3, 121.2, 111.7, 111.4, 109.5, 102.5, 55.8, 54.3, 52.3, 47.2, 39.7, 29.9, 21.4 ppm; HRMS (ESI^+) m/z calcd for $C_{36}H_{35}N_2O_5S$ $[M + H]^+$ 607.2267, found 607.2294.

*1-(1-Phenylvinyl)-2-tosyl-2,3,4,9-tetrahydro-1H-pyrido[3,4-b]-indole (**1ha**).* Yellow solid (17.1 mg, 40% yield) isolated using 10% ethyl acetate−petroleum ether (v/v); mp 164−166 °C; ^{1}H NMR ($CDCl_3$, 400 MHz) δ 7.72 (brs, 1H), 7.41−7.36 (m, 4H), 7.31−7.25 (m, 5H), 7.17−7.13 (m, 1H), 7.09−7.05 (m, 1H), 7.01 (d, J = 8 Hz, 2H), 6.23 (s, 1H), 5.36−5.36 (m, 1H), 5.06−5.05 (m, 1H), 3.82 (dd, J_1 = 14.4 Hz, J_2 = 6 Hz, 1H), 3.28−3.20 (m, 1H), 2.69−2.65 (m, 1H), 2.55 (dd, J_1 = 16.2 Hz, J_2 = 4.2 Hz, 1H), 2.27 (s, 3H); $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz) δ 147.5, 143.2, 139.5, 131.5, 129.3, 128.7, 128.5, 128.1, 127.6, 127.2, 126.8, 126.8, 122.4, 119.6, 118.4, 118.3, 111.0, 109.8, 56.0, 39.8, 21.5, 20.5 ppm; HRMS (ESI^+) m/z calcd for $C_{26}H_{25}N_2O_2S$ $[M+ H]^+$ 429.1637, found 429.1638.

*Methyl 4-(1-(2-Tosyl-2,3,4,9-tetrahydro-1H-pyrido[3,4-b]indol-1-yl)vinyl)benzoate (**1hb**).* Yellow solid (21.9 mg, 45% yield) isolated using 15% ethyl acetate−petroleum ether (v/v); mp 164−166 °C; ^{1}H NMR ($CDCl_3$, 400 MHz) δ 7.94−7.91 (m, 2H), 7.81 (brs, 1H), 7.41−7.28 (m, 6H), 7.19−7.15 (m, 1H), 7.09−7.05 (m, 1H), 7.00−6.98 (m, 2H), 6.19 (s, 1H), 5.43 (s, 1H), 5.11 (s, 1H), 3.92 (s, 3H), 3.85 (dd, J_1 = 14.6 Hz, J_2 = 5.8 Hz, 1H), 3.26−3.18 (m, 1H), 2.72−2.63 (m, 1H), 2.55 (dd, J_1 = 15.2 Hz, J_2 = 4.4 Hz, 1H), 2.26 (s, 3H); $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz) δ 166.9, 147.0, 144.0, 143.4, 137.4, 136.1, 129.8, 129.7, 129.3, 127.6, 127.1, 126.8, 122.5, 120.0, 119.8, 118.5, 111.1, 110.1, 56.0, 52.3, 39.7, 21.4, 20.4 ppm; HRMS (ESI^+) m/z calcd for $C_{28}H_{27}N_2O_4S$ $[M+ H]^+$ 487.1692, found 487.1696.

*1-(1-(4-Methoxyphenyl)vinyl)-2-tosyl-2,3,4,9-tetrahydro-1H-pyrido[3,4-b]indole (**1hh**).* Yellow solid (16.0 mg, 35% yield) isolated using 10% ethyl acetate−petroleum ether (v/v); mp 170−172 °C; ^{1}H NMR ($CDCl_3$, 400 MHz) δ 7.71 (brs, 1H), 7.43 (d, J = 8.4 Hz, 2H), 7.36 (d, J = 8.0 Hz, 1H), 7.28−7.20 (m, 3H), 7.15 (td, J_1 = 7.6 Hz, J_2 = 1.1 Hz, 1H), 7.08−7.01 (m, 3H), 6.84−6.80 (m, 2H), 6.17 (s, 1H), 5.31 (s, 1H), 5.00 (s, 1H), 3.87−3.81 (m, 4H), 3.29−3.22 (m, 1H), 2.70−2.62 (m, 1H), 2.54 (dd, J_1 = 15.2 Hz, J_2 = 4.4 Hz, 1H), 2.27 (s, 3H); $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz) δ 159.6, 146.8, 143.2, 137.6, 136.0, 131.8, 130.5, 129.3, 128.8, 127.2, 126.8, 122.3, 119.6, 118.4, 117.3, 113.9, 111.0, 109.7, 56.2, 55.4, 39.9, 21.5, 20.5 ppm; HRMS (ESI^+) m/z calcd for $C_{27}H_{27}N_2O_3S$ $[M+ H]^+$ 459.1742, found 459.1735.

*9-Butyl-1-(1-phenylvinyl)-2-tosyl-2,3,4,9-tetrahydro-1H-pyrido-[3,4-b]indole (**1ia**).* Pale yellow solid (36.8 mg, 76% yield) isolated using 10% ethyl acetate−petroleum ether (v/v); mp 98−100 °C; ^{1}H NMR ($CDCl_3$, 400 MHz) δ 7.76−7.61 (m, 4H), 7.45−7.30 (m, 3H), 7.31 (d, J = 8.0 Hz, 1H), 7.25−7.18 (m, 1H), 7.11−7.05 (m, 2H), 6.90 (d, J = 8.4 Hz, 2H), 6.33 (s, 1H), 5.48 (s, 1H), 4.72 (s, 1H), 4.11−3.92 (m, 2H), 3.76−3.71 (m, 1H), 3.39−3.31 (m, 1H), 2.84−2.75 (m, 1H), 2.59−2.54 (m, 1H), 2.23 (s, 3H), 1.92−1.77 (m, 2H), 1.46−1.41 (m, 2H), 0.99 (t, J = 7.2 Hz, 3H), 2.27 (s, 3H); $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz) δ 146.8, 143.4, 139.6, 137.3, 136.5, 134.9, 131.5, 130.6, 129.1, 128.5, 127.3, 125.5, 121.7, 119.9, 119.0, 118.5, 109.4, 108.7, 54.4, 43.5, 38.4, 32.5, 21.4, 20.6, 20.2, 14.0 ppm; HRMS (ESI^+) m/z calcd for $C_{30}H_{33}N_2O_2S$ $[M+ H]^+$ 485.2263, found 485.2281.

Transformation of 6a into THBC 1a (Scheme 2). To a solution of allenamide **6a** (0.1 mmol) in anhydrous THF (2 mL), *p*-TsOH (1.9 mg, 0.01 mmol) was added. The reaction mixture was stirred at room temperature. After completion of reaction (monitored by TLC), the solvents were removed under reduced pressure. Then the residue was purified by silica gel (100−200 mesh) column chromatography using 10% ethyl acetate−petroleum ether (v/v) used as eluting solvent to obtain the pure product **1a**.

*9-Methyl-2-tosyl-1-vinyl-2,3,4,9-tetrahydro-1H-pyrido[3,4-b]-indole (**1a**).* Brown gum (15.7 mg, 43% yield) isolated using 10% ethyl acetate−petroleum ether (v/v); ^{1}H NMR ($CDCl_3$, 400 MHz) δ 7.66−7.63 (m, 2H), 7.37 (dt, J_1 = 7.9 Hz, J_2 = 0.9 Hz, 1H), 7.28−7.24 (m, 1H), 7.21−7.17 (m, 1H), 7.12 (dd, J_1 = 8.4 Hz, J_2 = 0.8 Hz, 2H), 7.08−7.04 (m, 1H), 6.08−6.00 (m, 1H), 5.67 (d, J = 5.6 Hz, 1H), 5.31 (dt, J_1 = 7.6 Hz, J_2 = 1 Hz, 1H), 4.92 (dt, J_1 = 17.0 Hz, J_2 = 1.2

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Hz, 1H), 4.05−3.99 (m, 1H), 3.57 (s, 3H), 3.35−3.27 (m, 1H), 2.62−2.59 (m, 2H), 2.29 (s, 3H); $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz) δ 143.4, 138.1, 137.2, 134.9, 131.4, 129.6, 127.0, 126.4, 121.8, 119.9, 119.2, 118.3, 109.0, 108.2, 53.6, 39.0, 29.8, 21.5, 20.7 ppm; HRMS (ESI^+) m/z calcd for $C_{21}H_{23}N_2O_2S$ $[M+H]^+$ 367.1480, found 367.1478.

General Experimental Procedure for the Synthesis of β-Carbolines 2 (Table 4). To a well-stirred solution of **1** (0.05 mmol) in dry toluene (1.5 mL), DDQ (34.1 mg, 0.15 mmol) was added under argon atmosphere. Then reaction mixture was heated using an oil bath at 100 °C until completion (TLC). Then the solvent was evaporated *in vacuo*, and the reaction was quenched with a saturated solution of sodium bicarbonate. Then the resulting reaction mixture was extracted with ethyl acetate (3 × 20 mL). The combined organic extracts were then washed with brine, dried over anhydrous Na_2SO_4, and concentrated under reduced pressure. The product obtained was purified through silica gel column chromatography using 15−25% ethyl acetate−petroleum ether (v/v) as eluting solvent to obtain the pure β-carbolines **2**.

*9-Methyl-1-(1-phenylvinyl)-9H-pyrido[3,4-b]indole (**2a**).* Yellow gum (11.4 mg, 80% yield) isolated using 20% ethyl acetate−petroleum ether (v/v); ^1H NMR ($CDCl_3$, 400 MHz) δ 8.51 (d, J = 5.2 Hz, 1H), 8.16 (d, J = 7.6 Hz, 1H), 7.98 (d, J = 5.2 Hz, 1H), 7.60−7.56 (m, 1H), 7.38−7.24 (m, 7H), 6.19 (s, 1H), 5.55 (s, 1H), 3.72 (s, 3H); $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz) δ 147.3, 143.7, 142.5, 139.7, 138.4, 135.1, 129.9, 128.8, 128.6, 128.2, 126.7, 121.6, 121.1, 119.8, 117.4, 114.0, 109.6, 31.8 ppm; HRMS (ESI^+) m/z calcd for $C_{20}H_{17}N_2[M+H]^+$ 285.1392, found 285.1396.

*9-Methyl-1-(1-(p-tolyl)vinyl)-9H-pyrido[3,4-b]indole (**2b**).* Yellow gum (11.8 mg, 79% yield) isolated using 20% ethyl acetate−petroleum ether (v/v); ^1H NMR ($CDCl_3$, 600 MHz) δ 8.57 (d, J = 5.4 Hz, 1H), 8.21 (d, J = 7.8 Hz, 1H), 8.09−8.08 (m, 1H), 7.64 (t, J = 7.5 Hz, 1H), 7.41 (d, J = 8.4 Hz, 1H), 7.35 (t, J = 7.5 Hz, 1H), 7.20 (d, J = 7.8 Hz, 2H), 7.11 (d, J = 7.8 Hz, 2H), 6.24 (s, 1H), 5.62 (s, 1H), 3.73 (s, 3H), 2.33 (s, 3H); $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz) δ 143.0, 142.7, 138.5, 136.7, 136.5, 134.8, 130.8, 129.6, 129.3, 126.5, 121.9, 120.8, 120.3, 118.0, 114.4, 109.8, 31.9, 21.3 ppm; HRMS (ESI^+) m/z calcd for $C_{21}H_{19}N_2$ $[M+H]^+$ 299.1548, found 299.1548.

*1-(1-(4-Fluorophenyl)vinyl)-9-methyl-9H-pyrido[3,4-b]indole (**2c**).* Brown gum (12.1 mg, 80% yield) isolated using 20% ethyl acetate−petroleum ether (v/v); ^1H NMR ($CDCl_3$, 300 MHz) δ 8.53 (d, J = 5.1 Hz, 1H), 8.19 (d, J = 8.1 Hz, 1H), 8.02 (d, J = 5.4 Hz, 1H), 7.65−7.59 (m, 1H), 7.42−7.26 (m, 4H), 6.99 (t, J = 8.7 Hz, 2H), 6.16 (s, 1H), 5.56 (s, 1H), 3.74 (s, 3H); $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz) δ 162.8 (d, J = 247 Hz), 143.1, 142.7, 138.1, 135.7 (d, J = 3.4 Hz), 135.0, 130.4, 130.3, 128.9, 128.4 (d, J = 8.1 Hz), 121.7, 121.0, 120.1, 117.5, 115.7 (d, J = 21.5 Hz), 114.2, 109.7, 31.9 ppm; HRMS (ESI^+) m/z calcd for $C_{20}H_{16}N_2F$ $[M+H]^+$ 303.1298, found 303.1299.

*Methyl 4-(1-(9-Methyl-9H-pyrido[3,4-b]indol-1-yl)vinyl)benzoate (**2d**).* Yellow gum (14.5 mg, 85% yield) isolated using 25% ethyl acetate−petroleum ether (v/v); ^1H NMR ($CDCl_3$, 600 MHz) δ 8.54 (d, J = 5.4 Hz, 1H), 8.19 (d, J = 7.8 Hz, 1H), 8.03 (d, J = 5.4 Hz, 1H), 7.97 (d, J = 8.4 Hz, 2H), 7.62−7.60 (m, 1H), 7.42−7.38 (m, 3H), 7.32 (t, J = 7.2 Hz, 1H), 6.32 (s, 1H), 5.70 (s, 1H), 3.90 (s, 3H), 3.71 (s, 3H); $^{13}C\{^1H\}$ NMR ($CDCl_3$, 150 MHz) δ 166.7, 146.3, 143.8, 142.7, 142.4, 138.3, 135.0, 130.1, 130.0, 129.6, 128.7, 126.5, 121.5, 120.9, 119.9, 119.6, 114.2, 109.6, 52.1, 31.8 ppm; HRMS (ESI^+) m/z calcd for $C_{22}H_{19}N_2O_2$ $[M+H]^+$ 343.1447, found 343.1454.

*9-Benzyl-1-(1-phenylvinyl)-9H-pyrido[3,4-b]indole (**2e**).* Yellow gum (13.3 mg, 74% yield) isolated using 15% ethyl acetate−petroleum ether (v/v); ^1H NMR ($CDCl_3$, 400 MHz) δ 8.55 (d, J = 5.2 Hz, 1H), 8.22−8.20 (m, 1H), 8.05 (d, J = 5.2 Hz, 1H), 7.51−7.47 (m, 1H), 7.32−7.28 (m, 1H), 7.25−7.23 (m, 1H), 7.19 (m, 5H), 7.12−7.11 (m, 3H), 6.75−6.73 (m, 2H), 5.86 (s, 1H), 5.52 (s, 2H), 5.21 (s, 1H); $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz) δ 145.7, 143.7, 142.4, 138.9, 138.4, 137.4, 134.8, 130.5, 128.9, 128.7, 128.6, 128.2, 127.1, 126.4, 125.6, 121.7, 121.4, 120.4, 117.4, 114.2, 110.6, 47.9 ppm; HRMS (ESI^+) m/z calcd for $C_{26}H_{21}N_2[M+H]^+$ 361.1705, found 361.1706.

*9-Benzyl-1-(1-(4-methoxyphenyl)vinyl)-9H-pyrido[3,4-b]indole (**2f**).* Brown gum (12.3 mg, 63% yield) isolated using 15% ethyl acetate−petroleum ether (v/v); ^1H NMR ($CDCl_3$, 400 MHz) δ 8.66−8.65 (m, 1H), 8.28 (d, J = 8 Hz, 2H), 7.62 (t, J = 7.8 Hz, 1H), 7.41 (t, J = 7.4 Hz, 1H), 7.32 (d, J = 8.8 Hz, 1H), 7.15−7.09 (m, 3H), 7.01 (d, J = 8.4 Hz, 2H), 6.69−6.65 (m, 4H), 5.93 (s, 1H), 5.51 (s, 2H), 5.31 (s, 1H), 3.73 (s, 3H); $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz) δ 159.9, 146.0, 140.2, 136.9, 131.2, 128.6, 128.5, 127.7, 127.2, 127.1, 125.8, 125.5, 122.0, 121.0, 121.0, 114.6, 114.3, 114.2, 110.9, 100.0, 55.3, 48.0 ppm; HRMS (ESI^+) m/z calcd for $C_{27}H_{23}N_2O$ $[M+H]^+$ 391.1810, found 391.1812.

*Methyl 4-(1-(9-Allyl-9H-pyrido[3,4-b]indol-1-yl)vinyl)benzoate (**2g**).* Yellow gum (16.0 mg, 87% yield) isolated using 20% ethyl acetate−petroleum ether (v/v); ^1H NMR ($CDCl_3$, 600 MHz) δ 8.55 (d, J = 5.4 Hz, 1H), 8.20 (d, J = 7.8 Hz, 1H), 8.04 (d, J = 5.4 Hz, 1H), 7.96 (d, J = 8.4 Hz, 2H), 7.58 (t, J = 7.2 Hz, 1H), 7.40−7.36 (m, 3H), 7.33 (t, J = 7.5 Hz, 1H), 6.26 (s, 1H), 5.66 (s, 1H), 5.63−5.57 (m, 1H), 4.95 (d, J = 10.2 Hz, 1H), 4.89 (s, 2H), 4.72 (d, J = 17.4 Hz, 1H), 3.90 (s, 3H); $^{13}C\{^1H\}$ NMR ($CDCl_3$, 150 MHz) δ 166.8, 143.2, 142.6, 142.0, 138.4, 134.4, 132.2, 130.7, 130.0, 129.6, 128.7, 126.5, 121.5, 121.2, 120.2, 119.3, 116.4, 114.2, 110.5, 52.1, 46.7 ppm; HRMS (ESI^+) m/z calcd for $C_{24}H_{21}N_2O_2$ $[M+H]^+$ 369.1603, found 369.1599.

*6-Methoxy-9-methyl-4-phenyl-1-(1-phenylvinyl)-9H-pyrido[3,4-b]indole (**2h**).* Yellow solid (18.1 mg, 93% yield) isolated using 15% ethyl acetate−petroleum ether (v/v); mp 178−179 °C; ^1H NMR ($CDCl_3$, 400 MHz) δ 8.39 (s, 1H), 7.67−7.65 (m, 2H), 7.59−7.51 (m, 3H), 7.34−7.18 (m, 7H), 7.01 (m, 1H), 6.26 (s, 1H), 5.66 (s, 1H), 3.73 (s, 3H), 3.60 (s, 3H); $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz) δ 153.8, 150.5, 135.1, 131.7, 130.2, 129.6, 128.9, 128.9, 128.9, 128.8, 128.6, 128.6, 128.5, 126.6, 126.2, 125.9, 121.0, 120.8, 110.5, 105.4, 55.7, 32.1 ppm; HRMS (ESI^+) m/z calcd for $C_{27}H_{23}N_2O$ $[M+H]^+$ 391.1810, found 391.1801.

Experimental Procedure to Synthesize Product 8. To a well-stirred solution of **1aa** (44 mg, 0.1 mmol) in dry 1,4-dioxane (2 mL) was added KO^{t}Bu (22.4 mg, 0.2 mmol) under an argon atmosphere, and the reaction mixture was allowed to reflux at 110 °C using an oil bath. After completion of the reaction (TLC), the solvent was evaporated in vacuo. The resulting mixture was extracted with ethyl acetate (3 × 20 mL). The combined organic extracts were then washed with brine, dried over anhydrous Na_2SO_4, and concentrated under reduced pressure. The crude product was purified by column chromatography using 15% ethyl acetate−petroleum ether (v/v) as eluent to obtain the pure **8** as a white solid (38.0 mg, 86% yield) isolated using 15% ethyl acetate−petroleum ether (v/v): mp 130−132 °C; ^1H NMR ($CDCl_3$, 400 MHz) δ 7.25−7.22 (m, 7H), 7.05−7.01 (m, 2H), 6.94−6.91 (m, 2H), 6.54 (dd, J_1 = 8.4 Hz, J_2 = 0.4 Hz, 2H), 4.55−4.49 (m, 1H), 3.81−3.73 (m, 1H), 2.72−2.66 (m, 4H), 2.60−2.54 (m, 4H), 1.80 (s, 3H); $^{13}C\{^1H\}$ NMR ($CDCl_3$, 100 MHz) δ 143.1, 141.4, 139.7, 137.8, 136.2, 133.2, 128.8, 128.5, 128.3, 127.8, 126.6, 126.5, 125.0, 122.3, 119.2, 117.9, 109.4, 109.2, 47.2, 31.3, 21.3, 20.9, 19.5 ppm; HRMS (ESI^+) m/z calcd for $C_{27}H_{27}N_2O_2S$ $[M+H]^+$ 443.1793, found 443.1795.

Experimental Procedure for the Synthesis of Products 9 (Scheme 3). To a well-stirred DMF (2 mL) solution of $Pd(PPh_3)_4$ (5.8 mg, 5 mol %) and PCy_3 (5.9 mg, 12 mol %) was added aryl iodide (0.11 mmol) with stirring. After 30 min of stirring at rt, Cs_2CO_3 (130.4 mg, 0.4 mmol) and allenamide **6h** (0.1 mmol) were successively added. The whole reaction mixture was then heated using an oil bath to 100 °C for the requisite time. After completion of the reaction (monitored by TLC), the solvent was evaporated under reduced pressure; the resulting residue was mixed with water (10 mL) and extracted with ethyl acetate (3 × 10 mL). The combined organic extracts were washed with brine (10 mL), dried over anhydrous Na_2SO_4, and concentrated under vacuum. The crude product obtained was purified through silica gel (100−200 mesh) column chromatography using 30−50% ethyl acetate−petroleum ether (v/v) as eluent to obtain the pure the product **9** in moderate to good yields. In case of aryl iodides having EWG group, mostly one diastereomer was obtained, but aryl iodides having neutral or EDG group led to the

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generation of inseparable mixtures of diastereomers in the ratio of about 1.5:1 (from 1H NMR) for **9a** and 2.2:1 for **9g** (for details, see Figures S8 and S9).

Characterization of 9a,b/9i/9g. *2′-(1-Phenylvinyl)-1′-tosylspiro[indole-3,3′-pyrrolidine] (**9a**).* Brown solid (27.0 mg, 63% combined yield, dr 1.5:1) isolated using 30% ethyl acetate−petroleum ether (v/v); mp 128−130 °C; ^{1}H NMR ($CDCl_3$, 600 MHz) δ 7.87−7.84 (m, 4H), 7.63 (s, 1H), 7.47−7.36 (m, 6H), 7.29−7.27 (m, 1H), 7.23−7.18 (m, 7H), 7.07−6.98 (m, 5H), 6.76 (d, J = 7.2 Hz, 2H), 6.55 (d, J = 7.2 Hz, 1H), 5.59 (s, 1H), 5.56 (s, 1H), 5.54 (s, 1H), 5.37 (s, 1H), 5.02 (s, 1H), 4.98 (s, 1H), 3.98−3.77 (m, 4H), 2.65−2.60 (m, 1H), 2.51−2.50 (m, 6H), 2.28−2.24 (m, 1H), 1.90−1.86 (m, 1H), 1.59−1.57 (m, 1H); ^{13}C{^{1}H} NMR ($CDCl_3$, 150 MHz) δ 173.2, 172.6, 155.1, 154.2, 150.3, 147.2, 146.2, 144.3, 144.0, 141.2, 138.5, 135.9, 135.6, 134.4, 130.1, 129.9, 128.6, 128.5, 128.5, 128.3, 127.9, 127.6, 127.6, 127.6, 126.5, 126.4, 126.1, 126.0, 124.5, 121.4, 121.2, 121.0, 115.4, 66.9, 65.5, 65.4, 62.9, 48.1, 47.2, 32.0, 30.8, 21.7, 21.6 ppm; HRMS (ESI$^+$) m/z calcd for $C_{26}H_{25}N_2O_2S$ [M + H]$^+$429.1637, found 429.1655.

*Methyl 4-(1-(1′-Tosylspiro[indole-3,3′-pyrrolidin]-2′-yl)vinyl)-benzoate (**9b**).* Yellow gum (31.6 mg, 65% yield) isolated using 50% ethyl acetate−petroleum ether (v/v); ^{1}H NMR ($CDCl_3$, 600 MHz) δ 7.88−7.84 (m, 4H), 7.63 (s, 1H), 7.45−7.42 (m, 4H), 7.28−7.24 (m, 2H), 6.99 (t, J = 7.5 Hz, 1H), 6.51 (d, J = 7.8 Hz, 1H), 5.63 (s, 1H), 5.52 (s, 1H), 4.97 (s, 1H), 3.93−3.87 (m, 5H), 2.59−2.55 (m, 1H), 2.52 (s, 3H), 2.40−2.40 (m, 1H); ^{13}C{^{1}H} NMR ($CDCl_3$, 150 MHz) δ 172.3, 166.5, 154.1, 146.5, 144.1, 143.0, 140.9, 135.3, 130.0, 129.8, 129.8, 128.6, 127.6, 126.7, 126.4, 121.5, 121.2, 117.0, 66.8, 65.6, 52.1, 47.4, 30.9, 21.6 ppm; HRMS (ESI$^+$) m/z calcd for $C_{28}H_{26}N_2O_4SNa$ [M + Na]$^+$ 509.1511, found 509.1512.

*1-(4-(1-(1′-Tosylspiro[indole-3,3′-pyrrolidin]-2′-yl)vinyl)phenyl)-ethan-1-one (**9i**).* Yellow gum (28.9 mg, 61% yield) isolated using 35% ethyl acetate−petroleum ether (v/v); ^{1}H NMR ($CDCl_3$, 600 MHz) δ 7.89 (d, J = 7.8 Hz, 2H), 7.82 (s, 1H), 7.72 (d, J = 7.8 Hz, 1H),), 7.61 (d, J = 8.4 Hz, 1H), 7.43−7.37 (m, 3H), 7.32 (d, J = 8.4 Hz, 1H), 7.21−7.17 (m, 1H), 7.11−7.08 (m, 1H), 7.00 (d, J = 8.4 Hz, 1H), 6.23 (s, 1H), 5.48 (s, 1H), 5.14 (s, 1H), 3.86−3.83 (m, 1H), 3.40−3.38 (m, 1H), 3.26−3.21 (m, 1H), 2.62 (s, 3H), 2.57−2.54 (m, 1H), 2.27 (s, 3H); ^{13}C{^{1}H} NMR ($CDCl_3$, 100 MHz) δ 197.9, 170.9, 146.9, 144.2, 143.4, 137.3, 132.2, 132.1, 129.3, 128.7, 128.6, 127.8, 127.1, 120.3, 119.6, 118.4, 111.1, 100.0, 56.0, 39.6, 29.8, 26.7, 21.4, 20.2 ppm; HRMS (ESI$^+$) m/z calcd for $C_{28}H_{27}N_2O_3S$ [M + H]$^+$ 471.1742, found 471.1735.

*2′-(1-(p-Tolyl)vinyl)-1′-tosylspiro[indole-3,3′-pyrrolidine] (**9g**).* Yellow gum (24.8 mg, 56% combined yield, dr 2.2:1) isolated using 30% ethyl acetate−petroleum ether (v/v); ^{1}H NMR ($CDCl_3$, 600 MHz) δ 7.88−7.79 (m, 3H), 7.67−7.63 (m, 1H), 7.46−7.26 (m, 7H), 7.22−7.00 (m, 11H), 6.83−6.82 (m, 1H), 6.60−6.59 (m, 1H), 6.22 (s, 0.4 H), 5.51 (s, 1H), 5.34 (s, 0.4H), 5.29 (s, 1H), 5.01 (s, 0.4H), 4.98 (s, 1H), 3.91−3.83 (m, 2H), 3.30−3.25 (m, 1H), 2.94−2.92 (m, 0.4H), 2.72−2.63 (m, 1H), 2.58−2.55 (m, 1H), 2.51−2.46 (m, 4H), 2.45−2.27 (m, 6H); ^{13}C{^{1}H} NMR ($CDCl_3$, 150 MHz) δ 175.9, 175.9, 147.2, 146.9, 143.9, 143.1, 138.3, 137.8, 137.5, 136.5, 135.9, 135.7, 135.6, 130.4, 129.9, 129.6, 129.3, 129.1, 129.1, 128.5, 128.3, 128.2, 127.6, 127.4, 127.1, 127.0, 126.8, 126.7, 126.2, 122.1, 121.4, 121.3, 119.5, 119.4, 118.2, 117.7, 110.9, 109.5, 65.5, 55.9, 47.1, 39.7, 30.7, 21.6, 21.4, 21.2, 21.1 ppm; HRMS (ESI$^+$) m/z calcd for $C_{27}H_{27}N_2O_2S$ [M + H]$^+$ 443.1793, found 443.1792.

General Procedure to synthesize 10a, b and 11a,b (Scheme 5). A mixture of **2e−f** (0.05 mmol), $RuCl_3$ (0.5 mg, 0.0025 mmol), and $NaIO_4$ (64.2 mg, 0.3 mmol) in the solvent system CH_3CN/CCl_4/H_2O (2:1:1) was taken in a sealed tube. Then the resulting reaction mixture was stirred at rt for 1 h, and the progress of the reaction was monitored by TLC. After the completion of reaction, the reaction mixture was diluted with ethyl acetate (10 mL) and the reaction was quenched with aqueous sodium thiosulfate. Then reaction mixture was filtered through a plug of Celite. The filtrate was washed with water and brine. The combined ethyl acetate extracts were dried over anhydrous Na_2SO_4, and the solvent was removed under reduced pressure. The crude product was purified by column chromatography on silica gel (100−200 mesh) using 15% ethyl acetate−petroleum ether (v/v) as eluent to obtain the pure **10** in 40−45% yield. A mixture of 1-aroyl-β-carboline **10** (0.02 mmol) and anhydrous $AlCl_3$ (0.06 mmol) in dry toluene (1 mL) was stirred at rt for 5 h for the generation of reported useful products **11a**[27] and **11b**[28] in good yield (80−85%).

*(9-Benzyl-9H-pyrido[3,4-b]indol-1-yl)(phenyl)methanone (**10a**).* Yellow solid (8.2 mg, 45% yield) isolated using 15% ethyl acetate−petroleum ether (v/v); mp 148−149 °C;^{1}H NMR ($CDCl_3$, 400 MHz) δ 8.48 (d, J = 5.2 Hz, 1H), 8.25 (d, J = 8 Hz, 1H), 8.16 (d, J = 5.2 Hz, 1H), 7.67−7.63 (m, 1H), 7.58−7.54 (m, 3H), 7.46−7.36 (m, 2H), 7.28−7.25 (m, 2H), 6.88−6.81 (m, 3H), 6.64 (d, J = 7.2 Hz, 2H), 5.61 (s, 2H); ^{13}C{^{1}H} NMR ($CDCl_3$, 100 MHz) δ 194.0, 143.2, 140.6, 137.2, 136.6, 135.9, 134.4, 133.0, 130.7, 129.5, 128.5, 128.0, 127.4, 126.9, 121.8, 121.0, 120.7, 116.5, 110.2, 100.0, 48.1 ppm; HRMS (ESI$^+$) m/z calcd for $C_{25}H_{19}N_2O$ [M + H]$^+$ 363.1497, found 363.1496.

*(9-Benzyl-9H-pyrido[3,4-b]indol-1-yl)(4-methoxyphenyl)-methanone (**10b**).* Yellow solid (7.8 mg, 40% yield) isolated using 15% ethyl acetate−petroleum ether (v/v); mp 153−155 °C;^{1}H NMR ($CDCl_3$, 400 MHz) δ 8.50 (d, J = 4.8 Hz, 1H), 8.24 (d, J = 7.6 Hz, 1H), 8.15 (d, J = 5.2 Hz, 1H), 7.66−7.53 (m, 4H), 7.39−7.35 (,m, 1H), 6.90−6.84 (m, 3H), 6.76−6.73 (m, 2H), 6.67−6.65 (m, 2H), 5.59 (s, 2H), 3.82 (s, 3H); ^{13}C{^{1}H} NMR ($CDCl_3$, 100 MHz) δ 193.7, 163.7, 143.2, 137.1, 136.1, 134.3, 133.2, 132.0, 129.6, 129.5, 128.4, 127.5, 127.4, 126.8, 121.8, 121.0, 120.7, 116.2, 113.5, 110.2, 55.6, 48.1 ppm; HRMS (ESI$^+$) m/z calcd for $C_{26}H_{21}N_2O_2$[M + H]$^+$ 393.1603, found 393.1599.

*Phenyl(9H-pyrido[3,4-b]indol-1-yl)methanone[27] (**11a**).* Yellow solid (4.6 mg, 85% yield) isolated using 20% ethyl acetate−petroleum ether (v/v); mp 131−132 °C;^{1}H NMR ($CDCl_3$, 400 MHz) δ 10.48 (s, 1H), 8.73 (s, 1H), 8.32−8.19 (m, 4H), 7.70−7.64 (m, 2H), 7.57 (t, J = 7.4 Hz, 2H), 7.41 (t, J = 7.4 Hz, 2H); HRMS (ESI$^+$) m/z calcd for $C_{18}H_{13}N_2O$ [M + H]$^+$ 273.1028, found 273.1029.

*(4-Methoxyphenyl)(9H-pyrido[3,4-b]indol-1-yl)methanone[28] (**11b**).* Yellow solid (4.8 mg, 80% yield) isolated using 20% ethyl acetate−petroleum ether (v/v); mp 167−168 °C; ^{1}H NMR ($CDCl_3$, 400 MHz) δ 11.94 (s, 1H), 8.49 (d, J = 5.2 Hz, 1H), 8.40 (d, J = 5.2 Hz, 1H), 8.30−8.28 (m, 3H), 7.75 (d, J = 8.0 Hz, 1H), 7.56 (t, J = 8.4 Hz, 1H), 7.27 (t, J = 8.0 Hz, 1H), 6.98−6.93 (m, 2H), 3.85 (s, 3H); ^{13}C{^{1}H} NMR (DMSO-d_6, 150 MHz) δ 185.6, 163.4, 141.8, 133.9, 131.4, 129.7, 129.4, 129.0, 128.7, 127.0, 126.2, 122.3, 120.6, 113.9, 113.4, 111.8, 56.0; HRMS (ESI$^+$) m/z calcd for $C_{19}H_{15}N_2O_2$[M + H]$^+$ 303.1134, found 303.1126.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.0c02651.

X-ray crystallographic data (ORTEP diagrams) of products **6a**, **1aa**, **1ah**, and **1db** along with their important crystal data, experimental procedures and spectral data of starting substrates, and ^{1}H and ^{13}C NMR spectra for all compounds synthesized (PDF)

Accession Codes

CCDC 2031343−2031346 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

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